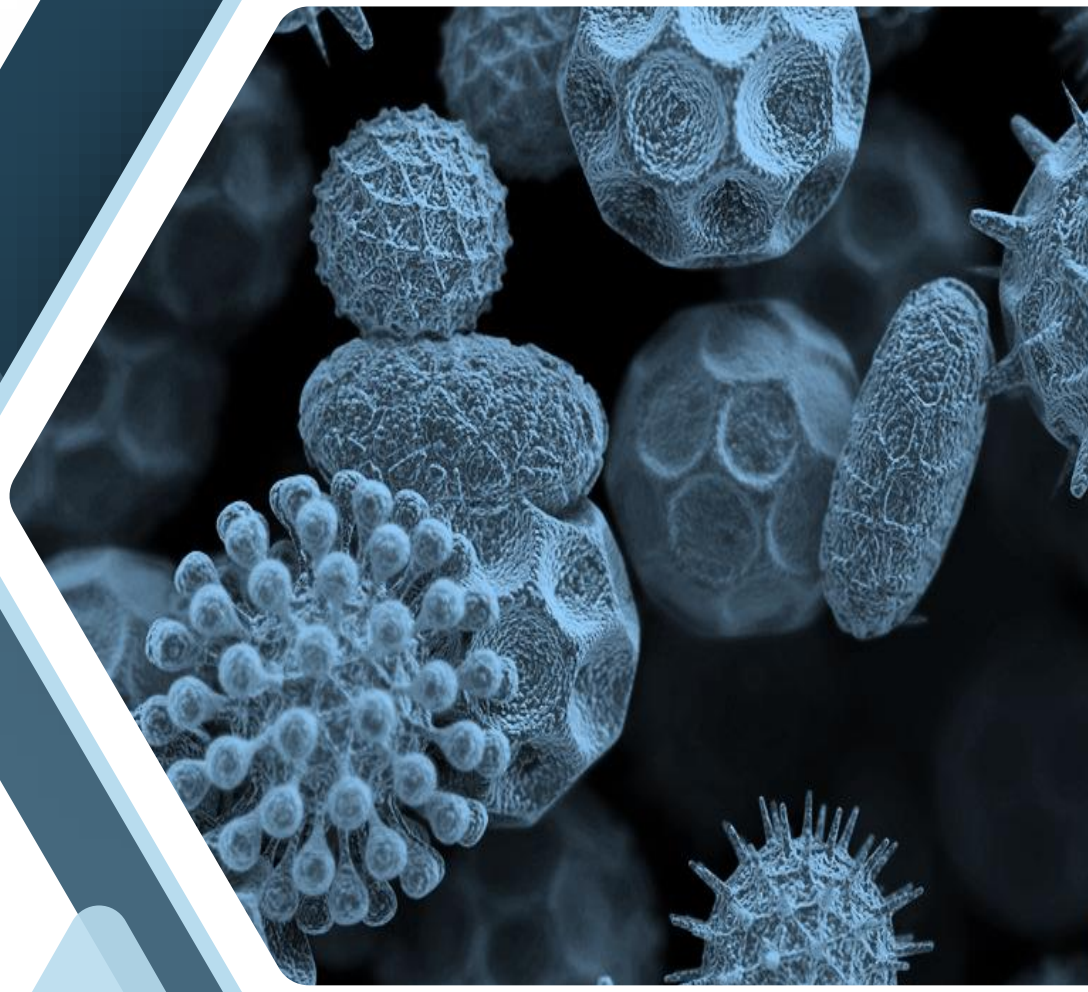
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CELL THERAPY HIGHLIGHTS Within the CGT Insights Corner 2025



Month: April 2026

Presenter: Octavus Team



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- The Opportunity 05
- The Science & Evolution 08
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- Ecosystem, Company Journey, & Outlook 31

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

PRIOR ISSUES

- Highlights from AACR-25 (**May, 25**)
- US Tariff impact on Pharma industries (**Jun ,26)**
- EULAR-25: Key insights of Psoriatic Arthritis (**Jul, 25**)
- Antibody Drug Conjugate: The next big investment in Oncology (**Aug, 25**)
- Animal Health (**Oct ,25)**
- Obesity deep dive (**Nov ,25)**
- GLP-1 and GIP (**Dec, 25**)
- Hematological Highlights from ASH 2025 (**Jan, 26**)
- China Biotech's Global Positioning from JPM-26 (**Feb,26**)
- BIOSIMILARS RISING: The Catalyst For A New Age In Biologics Therapies (**Mar,26**)

Additional: 1 Beautiful Bill

INSIGHTS CORNER

Executive Summary

Part 1 of the CGT landscape, with primary focus on cell therapy

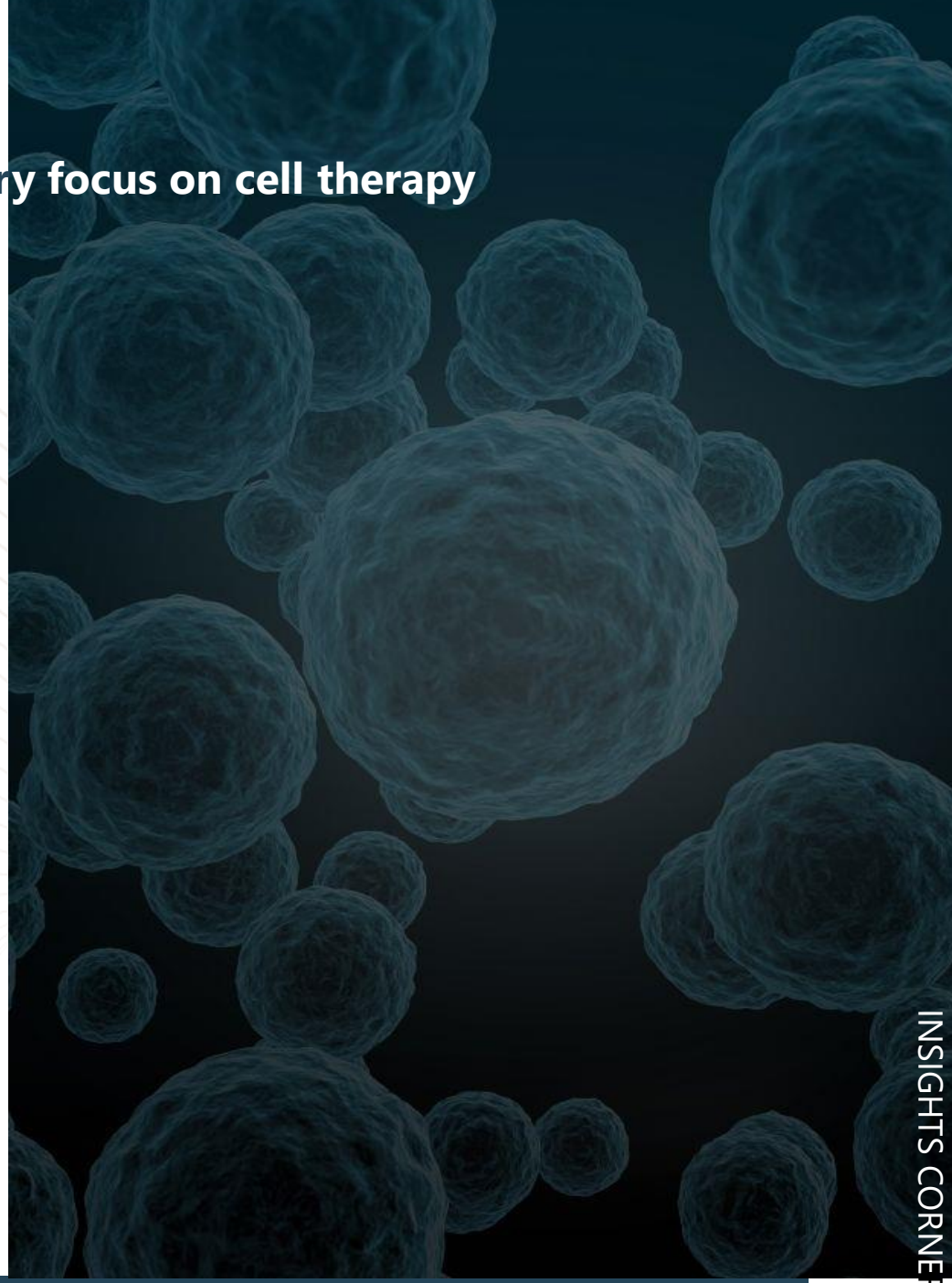
From niche to platform: Cell therapy is emerging as a mainstream biopharma growth area, supported by strong clinical, regulatory, and commercial momentum

CAR-T remains the dominant modality: The field is still CAR-T-led, while the technology landscape is broadening through TCR-T, CAR-NK, TIL therapies, and allogeneic/off-the-shelf approaches

2026 as a catalyst year: Near-term value creation in 2026 will likely be shaped by pipeline milestones, designation activity, and evolving global regulatory frameworks

Execution remains the bottleneck: The biggest constraints remain trial execution, manufacturing complexity, solid-tumor biology, reimbursement pressure, and long-term evidence generation

Scale and access will define winners: Companies that combine innovation with scalable access, strong partnerships, and differentiated product strategy are best positioned to lead the next phase of growth



INSIGHTS CORNER



The Opportunity: Why cell therapy is gaining momentum now

Why Cell Therapy is Trending?

1

Curative and regenerative potential (beyond oncology)

- Cell therapies can potentially treat diseases with no known cure by repairing or replacing damaged tissue, including conditions like Vision loss, age-related Macular Degeneration and multiple sclerosis (MS)
- Unlike treatments that only slow progression, early clinical data in MS, eye diseases, and Parkinson's suggest that stem cell derived therapies may restore function, driving increased interest in these areas

2

Rapid growth in oncology, with a shift toward solid tumors

- Global oncology incidence is rising steeply, and many patients still relapse or become resistant to current drugs, creating strong demand for new modalities like CAR-T and other cell therapies
- Since 2020, the number of CAR-T trials in solid tumors has increased substantially, reflecting growing sponsor efforts to translate CAR-T success beyond hematologic malignancies, despite the higher biological and clinical risks associated with solid tumor targets and microenvironments

3

Explosive increase in clinical trial activity overall

- Since around 2017, the number of registered global industry-sponsored cell therapy trials has surged, with ~5,000 trials now listed across clinical trial database, and this activity is expected to continue rising over the next decade as innovation and investment expand across a broad range of therapeutic areas and indications
- This volume signals that cell therapy is no longer a niche field; it is becoming a mainstream R&D pillar for big pharma, biotech's, and academic centers across many disease areas

4

Manufacturing and scalability are unlocking the next growth wave

- As trials and approvals grow, the focus is shifting from "can cell therapy work" to "how do we industrialize it?", accelerating investment in scalable, reliable and more affordable manufacturing solutions
- Advances in automation, allogeneic platforms and specialized CDMOs are rapidly transforming today's high-cost, manual and fragile process into standardized manufacturing platforms that can support broad patient access and global expansion of cell therapies

5

Big-pharma partnerships are validating cell therapy

- Big pharma and strategic partnerships are increasingly confirming the cell therapy space as a long-term, investable modality. Large pharmaceutical companies are entering multi-billion-dollar licensing deals, co-development agreements, and M&A transactions with specialized cell and gene therapy companies. These collaborations bring industrial-scale manufacturing, global commercial infrastructure, and late-stage clinical development ability that smaller players often lack. The steady flow of alliances and acquisitions signals that cell therapy is no longer viewed as an experimental niche but as a core pillar of future oncology and regenerative medicine portfolios

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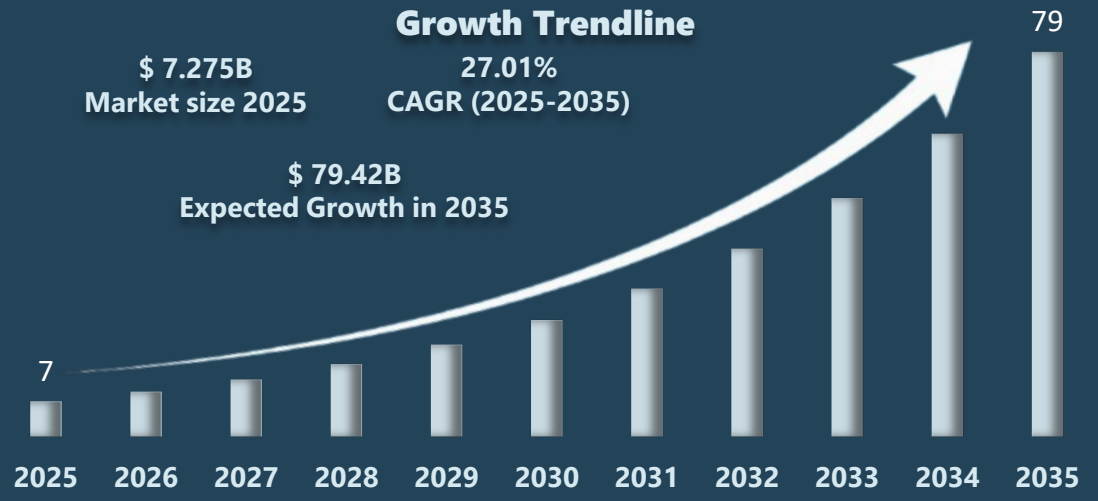
Cell Therapy Market-Key Stats

Growth Trendline

$ 7.275B
Market size 2025

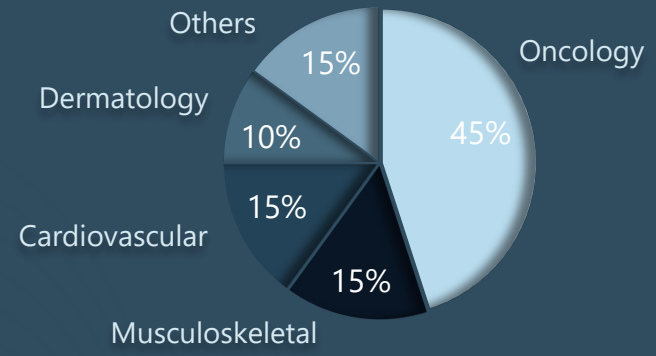
27.01%
CAGR (2025-2035)

$ 79.42B
Expected Growth in 2035



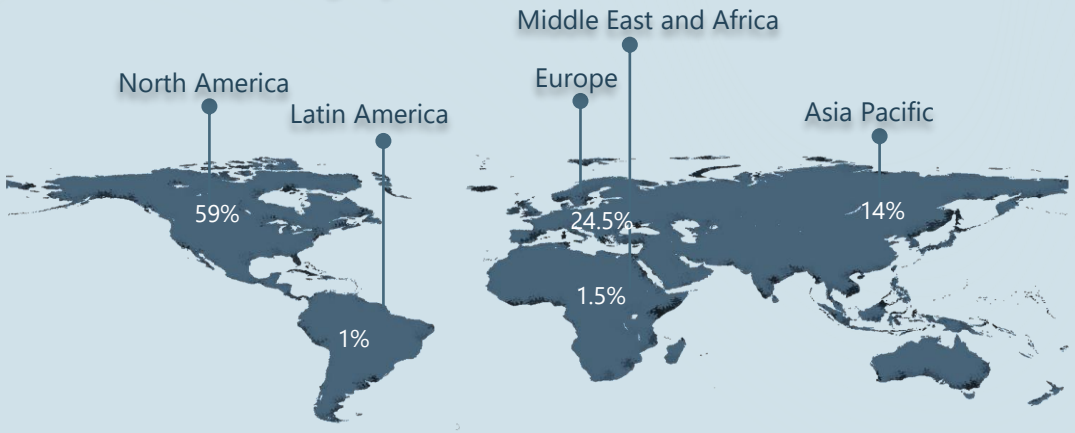
Note: Values shown in the graph are estimated and represent approximate figures

Therapy Landscape: Market Share Analysis (2025)



Other therapy areas include neurological disorders (e.g., Parkinson's, Alzheimer's and spinal cord injuries), autoimmune, ophthalmic, gastrointestinal, and rare genetic diseases.

Geographic Revenue Breakdown



Percentages show each region's relative dominance in the global cell therapy market.

Key Drivers

Asia-Pacific Expansion - Expansion of research and development in biopharma, rising investments in advanced cell and gene therapy framework, rapid growth in clinical trial are driving Asia Pacific as a high-growth region.

Clinical Pipeline - The cell and gene therapy is at a turning point, 900 CAR-T clinical trials started since 2020. From January to March 2025 there are 57% of the 79 new gene, cell, and RNA therapy trials were in oncology.

Rising Occurrence of Chronic & Rare Diseases - The increasing prevalence of age-related diseases, chronic conditions, cancer, autoimmune disorders, and neurological disorders has intensified demand for effective cell-based treatments**.**

Expanding Regulatory Approvals & Accelerated Pathways - Nearly 35% of regulatory approvals are linked to accelerated pathways, strengthening the market's growth trajectory. Regulatory acceleration programs are a significant factor influencing regional growth across the globe.

Limitations of Traditional Treatments - Cell therapies provide an alternative to traditional organ transplants, which face severe limitations like donor shortages and rejection risks.

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The Science & Evolution: Modalities, CAR-T evolution, and major milestones

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TYPES OF EMERGING CELL THERAPY TECHNOLOGIES

CAR-T CELL THERAPY

CAR T-cell therapy is an immunotherapy in which T cells are genetically modified to recognize and destroy cancer cells, mainly for certain blood cancers. It includes:
Autologous CAR-T: which uses the patient's own engineered T cells with lower rejection risk
Allogeneic CAR-T: which uses donor-derived "off-the-shelf" T cells that enable faster treatment but may carry risks of immune rejection or graft-versus-host reactions.

mRNA-BASED CELL THERAPY

mRNA-based cell therapy delivers mRNA into cells to produce therapeutic proteins, enabling temporary expression without altering DNA and offering a safer alternative to viral gene delivery in therapies like CAR-T.

CAR-NK CELL THERAPY

CAR-NK therapy is a cancer immunotherapy where natural killer (NK) cells are genetically engineered with CARs to target tumor antigens, enhancing cancer cell killing. These cells can come from donors, cord blood, or stem cells, enabling off-the-shelf treatment with lower risks of cytokine release syndrome and graft-versus-host disease than CAR-T therapy.

TCR-T CELL THERAPY

TCR-T cell therapy is a cancer immunotherapy in which T cells are genetically engineered with high-affinity T-cell receptors (TCRs) to recognize tumor antigens presented by MHC molecules, enabling targeted immune destruction of cancer cells.

TIL THERAPY

Tumor-infiltrating lymphocyte (TIL) therapy is an adoptive cell immunotherapy where T cells isolated from a patient's tumor are expanded in the laboratory and reinfused, often after lymphodepleting chemotherapy and with IL-2 support, to attack cancer cells.

IN VIVO CAR-T GENERATION

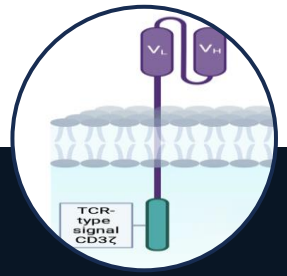
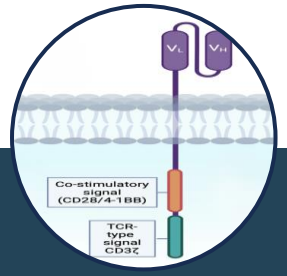
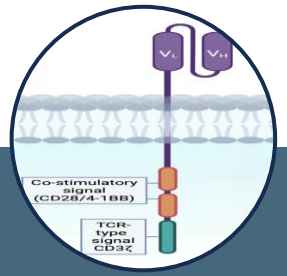
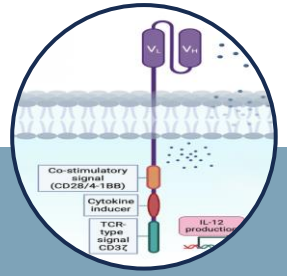
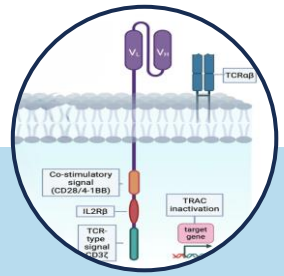
In vivo CAR-T generation delivers CAR-encoding genetic material directly into a patient's T cells inside the body, programming them to become CAR-T cells and eliminating the need for ex vivo cell collection and engineering.

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Evolution of CAR-T: From 1st, 2nd to Next Generation



CAR T-Cell Therapy

	 1st Generation	 2nd Generation	 3rd Generation	 4th Generation	 5th Generation
Summary	• Earliest CAR design: scFv + CD3ζ (or FcγR) signaling domain • First proof that T cells can be redirected to tumor antigens	• Addition of one co-stimulatory domain (CD28 or 4-1BB) + CD3ζ • Foundation of currently approved CAR-T therapies	• Incorporate two co-stimulatory domains (e.g., CD28 + 4-1BB or OX40)	• Armored CAR-T cells, that secrete cytokines upon activation • Add a third signal (cytokine signaling)	• Advanced CARs with built-in cytokine receptor signaling (JAK-STAT pathway) • Designed to mimic natural T cell activation fully
Working Mechanism	• Recognize tumor antigen via scFv • Activate T cells through CD3ζ signaling only • No co-stimulation, incomplete activation	• Provide two signals: • Antigen recognition (CD3ζ) • Co-stimulation (CD28 or 4-1BB) • Improves activation, proliferation, and survival	• Combine signaling pathways to: • Enhance activation • Improve persistence • Reduce exhaustion	• Recognize tumor, activate and release cytokines (e.g. IL-2, IL-12, etc) • Cytokines: • Enhance CAR-T function • Remodel tumor microenvironment • Recruit other immune cells	• Provide three integrated signals: • CD3ζ → activation • Co-stimulation (e.g., CD28) • Cytokine signaling (via JAK-STAT pathway) • No need for external cytokines
Impact	• First to redirect T cell specificity to tumor antigens, proving concept in preclinical trials	• Stronger T cell activation • Better persistence and expansion • Major clinical success (e.g., CD19-targeted therapies) • Approved therapies for leukemia and lymphoma (ABECMA, BREYANZI, etc)	• Potentially stronger and more durable responses • Better expansion and persistence in some studies • Synergistic signaling effects • Clinical studies is ongoing for lymphoma (HD-CAR-1, WZTL-002, etc)	• Overcome immunosuppressive tumor microenvironment • Improved activity in solid tumors • Promote antigen spreading and broader immunity • Enhanced persistence and proliferation	• Continuous, controlled cytokine signaling • Reduced T cell exhaustion • Better persistence and expansion • Improved tumor infiltration (especially solid tumors) • More physiological
Challenges	• No co-stimulatory signal • Poor *in vivo* persistence • Weak and short-lived antitumor response • Limited clinical success • Risk of toxicity • No approved CAR-T	• Toxicities: CRS (Cytokine release syndrome), Immune effector cell-associated neurotoxicity syndrome • Limited efficacy in solid tumor • Still risk of relapse (antigen loss)	• Limited clinical evidence of superiority over 2G • No consistent improvement in durability • Tumor escape still possible (lack of antigen spreading) • High cost and complexity	• High risk of systemic toxicity (CRS, off-tumor effects) • Difficult control of cytokine release • Preclinical success not fully predictive of human safety • Complex engineering	• Still in early/preclinical stages • Safety concerns (CRS, cytokine overactivation) • Need for optimized dosing and regulation • Limited clinical validation

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Cell Therapy Timeline

1988 — TIL Therapy Introduced

Adoptive transfer of tumor-infiltrating lymphocytes (TILs) is used to treat metastatic melanoma, demonstrating early success of T-cell based immunotherapy

1993 — 1st Generation CAR-T Cells

The first chimeric antigen receptor (CAR) T cells are developed, enabling engineered T cells to recognize tumor antigens

1998 — 2nd Generation CAR-T (CD28)

Addition of the CD28 co-stimulatory domain improves T-cell activation, proliferation, and antitumor activity

2004 — 2nd Generation CAR-T (4-1BB)

Incorporation of the 4-1BB co-stimulatory domain enhances persistence and memory formation of CAR-T cells

2006 — TCR-T Therapy Development

T-cell receptor engineered therapies show potential in treating cancers such as metastatic melanoma

2010 — 1st CD19 CAR-T Patient

Clinical trials showed successful leukemia treatment with CD19-targeted CAR-T cells, leading to the development of 3rd generation CAR-T therapies

2012 — 4th Generation CAR-T

"Armored" CAR-T cells are engineered to secrete cytokines to enhance immune responses

2017 — 1st CAR-T Approvals

The FDA approves the first CAR-T therapy for lymphoma patients

2021 — Expansion to Autoimmune Diseases

CAR-T therapy begins to be explored for autoimmune disorders such as systemic lupus erythematosus (SLE)

2023 — Next-Generation Engineering

Advanced CAR-T modifications are being developed to improve effectiveness, particularly against solid tumors

2024 — Major cell therapy milestones

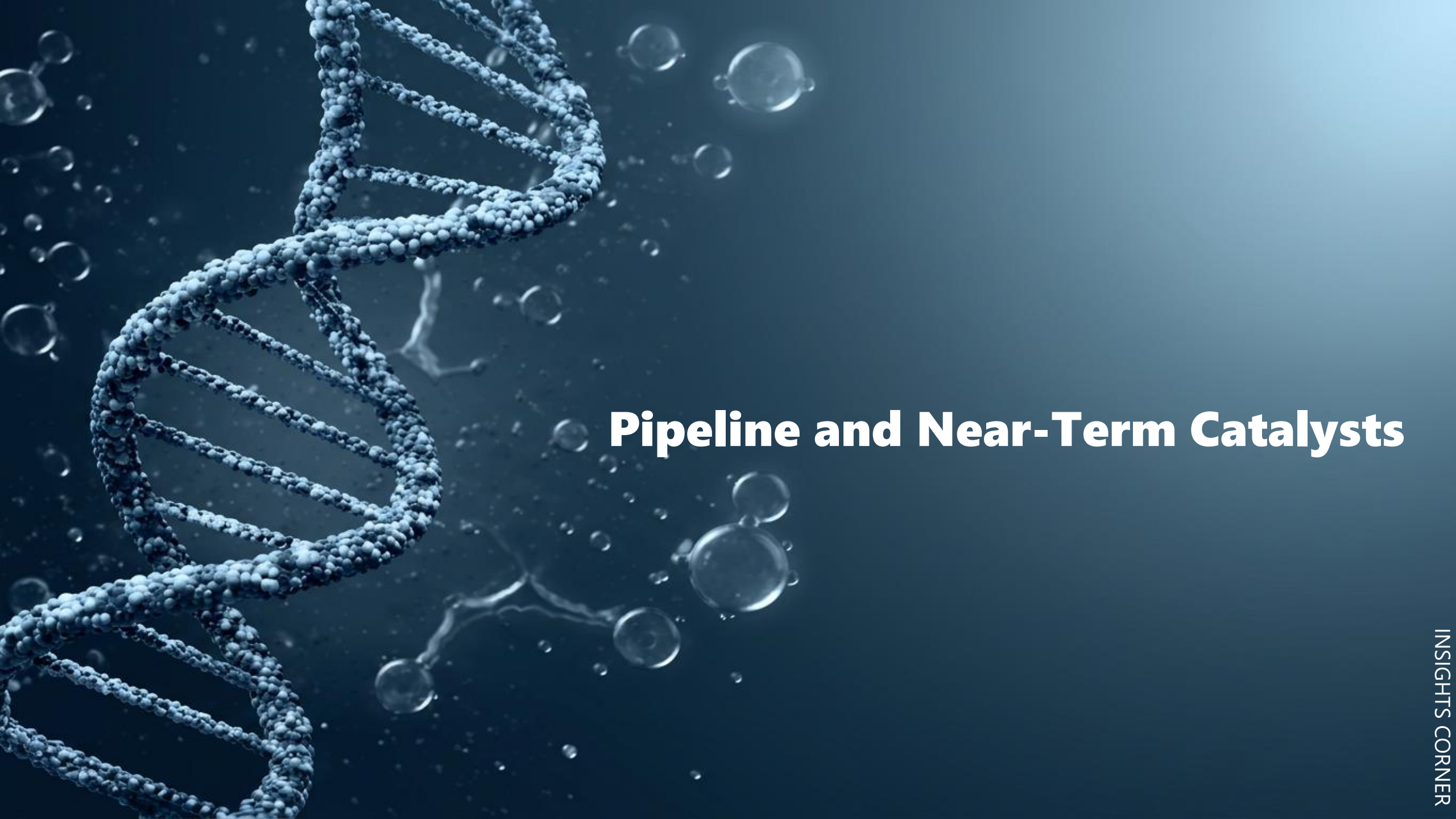
The US FDA approved Lifileucel, the 1st TIL therapy for unresectable or metastatic melanoma, and Afamitresgene autoleucel, the 1st TCR-T therapy for a solid tumor (synovial sarcoma). Later, Obecabtagene autoleucel became the 7th approved CAR-T therapy for R/R B-cell ALL, while new CAR-T trials in AML targeting CD33, CD123, and CLL1 reported encouraging remission outcomes

2025 — *In vivo* CAR-T cell engineering

In vivo CAR-T engineering enables direct generation of CAR-T cells within patients, eliminating complex ex vivo manufacturing while maintaining strong and sustained antitumor activity. Early ASCO 2025, P1 data show promising signals in solid tumors, including B7H3-CAR-T in glioblastoma and GCC19CART in metastatic colorectal cancer (80% ORR at higher doses)

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Pipeline and Near-Term Catalysts

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Approval at a Glance

2010-2015

- 2010 — PROVENGE (sipuleucel-T) — Dendreon
- 2011 — LaVIV (azficel-T) — FIBROCELL
- 2013 — Gintuit — ORGANOGENESIS
- 2015 — HOLOCLAR — HOLOSTEM

2016-2021

- 2016 — maci — VERICEL
- 2017 — spherOX — CODON
- 2017 — KYMRIAH (tisagenlecleucel) — NOVARTIS
- 2018 — KYMRIAH (tisagenlecleucel) — NOVARTIS
- 2017 — YESCARTA (axicabtagene ciloleucel) — Kite, A GILEAD Company
- 2018 — YESCARTA (axicabtagene ciloleucel) — Kite, A GILEAD Company
- 2020 — TECARTUS (brexucabtagene autoleucel) — Kite, A GILEAD Company
- 2020 — TECARTUS (brexucabtagene autoleucel) — Kite, A GILEAD Company
- 2021 — Breyanzi — Bristol Myers Squibb
- 2022 — Breyanzi — Bristol Myers Squibb
- 2021 — Abecma (idecabtagene vicleucel) — Bristol Myers Squibb
- 2021 — Abecma (idecabtagene vicleucel) — Bristol Myers Squibb
- 2021 — StrataGraft — Mallinckrodt
- 2021 — RETHYMIC — Sumitomo Pharma, Innovation today, healthier tomorrows

2022 - 2025

- 2022 — CARVYKTI (ciltacabtagene autoleucel) — Johnson&Johnson
- 2022 — CARVYKTI (ciltacabtagene autoleucel) — Johnson&Johnson
- 2022 — zynteglo (betibeglogene autotemcel) — GENETIX
- 2022 — skysona (elivaldogene autotemcel) — GENETIX
- 2022 — EBVALLO tabelecleucel — Pierre Fabre
- 2023 — Omisirge (omidubicel-onlv) — gamida cell
- 2023 — Lantidra — CellTrans
- 2023 — casgevy (exagamglogene autotemcel) — VERTEX
- 2024 — casgevy (exagamglogene autotemcel) — VERTEX
- 2023 — lyfgenia (lovotibeglogene autotemcel) — GENETIX
- 2024 — AMTAGVI (lifileucel) — IOVANCE BIOTHERAPEUTICS
- 2024 — lenmeldy (atidarsagene autotemcel) — libmeldy (atidarsagene autotemcel) — Orchard therapeutics
- 2020 — lenmeldy (atidarsagene autotemcel) — libmeldy (atidarsagene autotemcel) — Orchard therapeutics
- 2024 — Tecelra (afamitresgene autoleucel) — Adaptimmune
- 2024 — AUCATZYL (obecabtagene autoleucel) — Autolus
- 2025 — AUCATZYL (obecabtagene autoleucel) — Autolus
- 2024 — REGENCYTE — StemCyte
- 2024 — Ryoncil (remestemcel-L-rknd) — mesoblast
- 2025 — ENCELTO (revakinagene taroretcel-lwey) — neurotech
- 2025 — zevaskyn (prademagene zamikeracel) — Abeona THERAPEUTICS
- 2025 — WASKYRA — CORDEX
- 2025 — ZEMCELPRO — Telethon

CAR-T (Chimeric Antigen Receptor-T cells)

TCR-T (T-Cell Receptor Therapy)

TIL-T (Tumor-Infiltrating Lymphocyte Therapy)

Other Cell Therapy/ CGT

US EU

Intravenous Intradermal Topical Implant Intraarticular Intravitreal

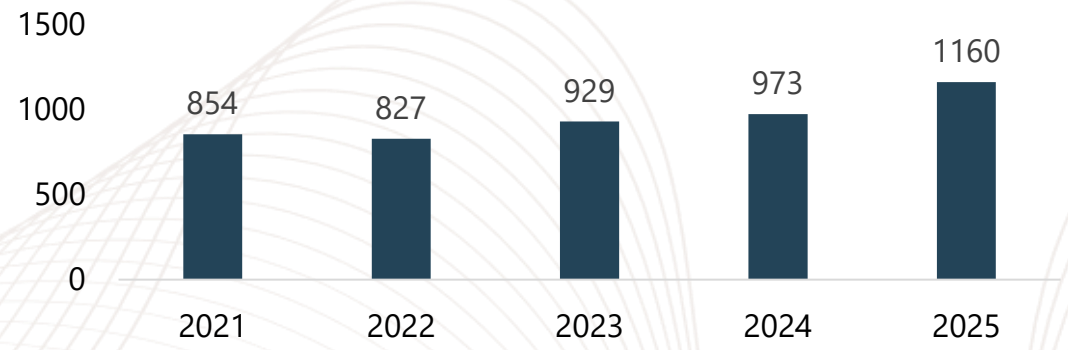
Note: Timeline ranges are based on first approval dates across US & EU, although BREYANZI received its first US approval in 2021 & EU approval in 2022, but placed within timeline range (2016–2021) based on the initial approval

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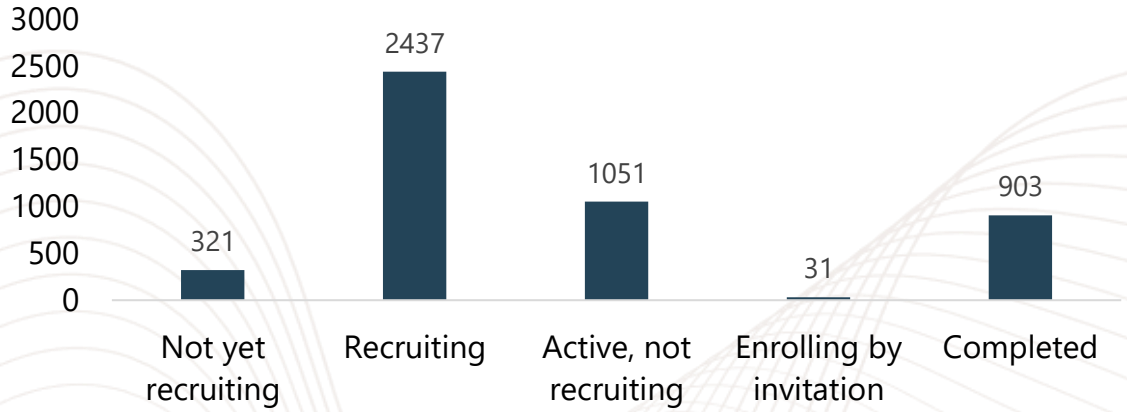
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Cell Therapy: Clinical Trials Overview

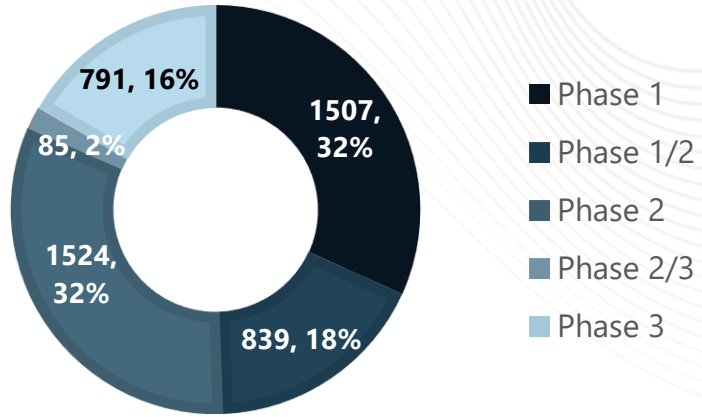
Number of Trials by Year



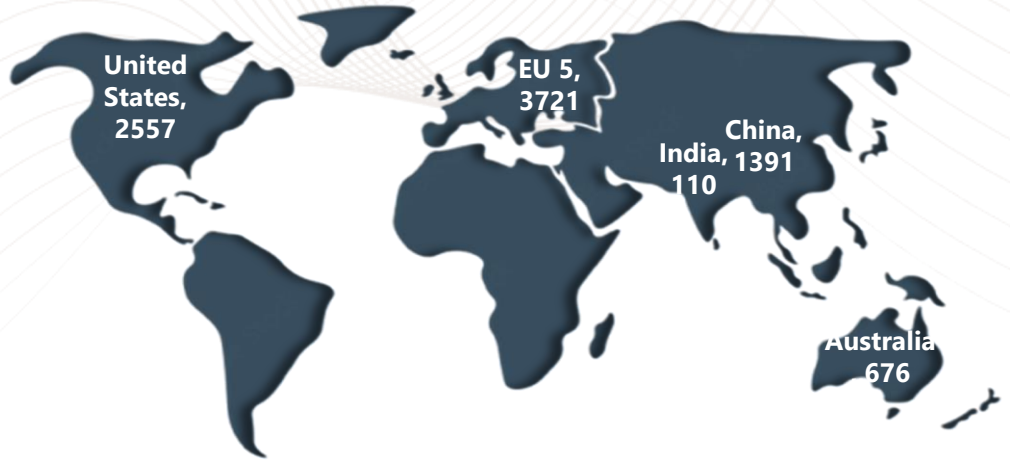
Trial Status



Number of Trials by Phase



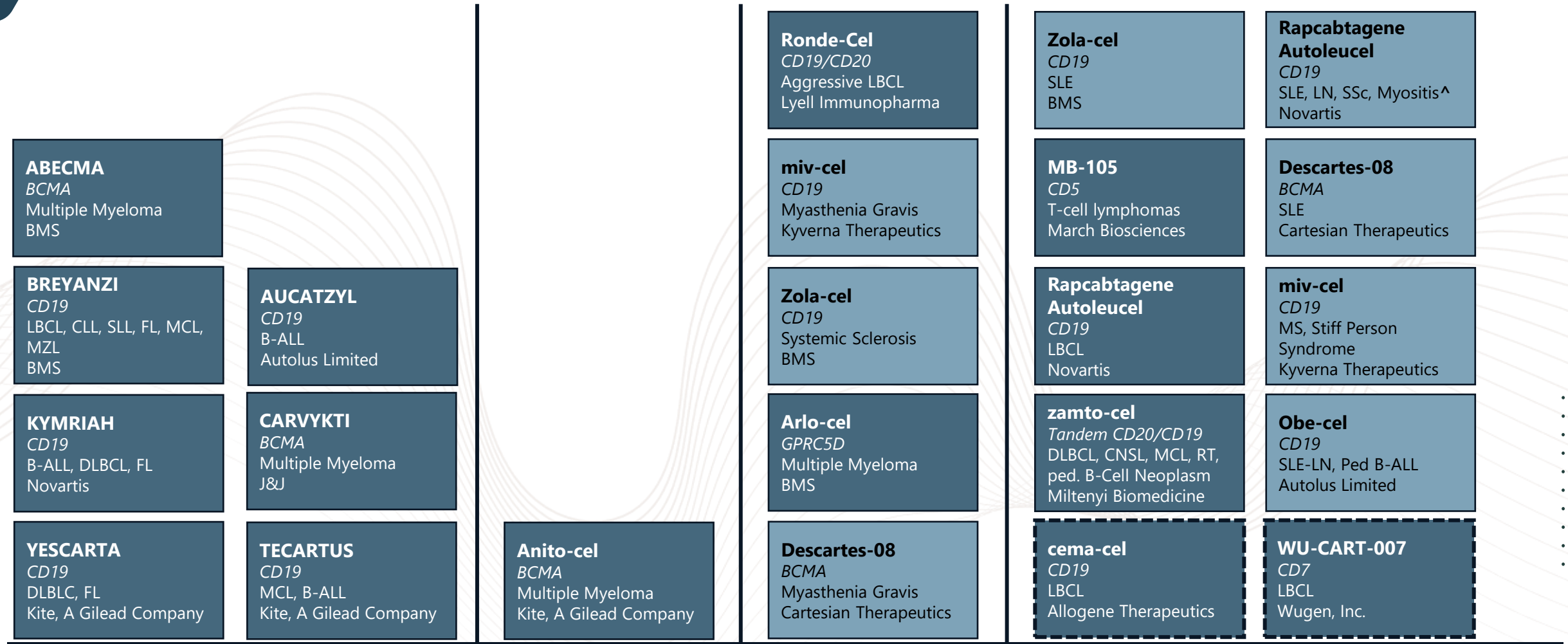
Trials by Geography



Note: Data has been collected for 5 years (Jan 2021 – Dec 2025) and covers interventional industry studies; EU 5 includes France, Spain, Germany, Italy and UK

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CAR-T's Landscape (1/2)



Note: Assets in development in the US and EU
^ANCA associated vasculitis

Immunology | Hem malignancies | Solid Tumors

Allogeneic
Autologous

Abbreviations: DLBCL: Diffuse Large B-Cell Lymphoma (LBCL); CNSL: Central Nervous System Lymphoma; MCL: Mantle Cell Lymphoma; RT: Richter Transformation; SLE: Systemic Lupus Erythematosus; SSc: Systemic Sclerosis; LN: Lupus Nephritis; B-ALL: B-cell Acute Lymphoblastic Leukemia; CLL: Chronic Lymphocytic Leukemia; SLL: Small Lymphocytic Leukemia; FL: Follicular Lymphoma; MZL: Marginal Zone Lymphoma

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CAR-T's Landscape (2/2)

			Zevor-cel/CT053 BCMA MM CARsgen Therapeutics	**CCTx-001** *IL-1RAP* AML Advesya SAS	**A2B694** *MSLN* Solid Tumors A2 Biotherapeutics	**A2B395** *EGFR* Solid Tumors A2 Biotherapeutics
	NXC-201 *BCMA* AL amyloidosis Immix Biopharma	**KITE-753** *CD19/CD20* B-cell Lymphoma Kite, A Gilead Company	**CT071** GPRC5D MM, RRpPCL CARsgen Therapeutics	**zugo-cel/CTX112** CD19 B-Cell Malignancies CRISPR Therapeutics	**IVS-3001** HLA-G Solid Tumors Invectys	**A2B530** *CEA* Solid Tumors A2 Biotherapeutics
QEL-005 *CD19* Refractory RA and SSc Quell Therapeutics	**zamto-cel** *Tandem CD20/CD19* Multiple Sclerosis Miltenyi Biomedicine	**KITE-363** *CD19/CD20* B-cell Lymphoma Kite, A Gilead Company	**CTD402** CD7 T-ALL, T-LBL Bioheng Therapeutics	**GLPG5101** *CD19* B cell malignancies Galapagos	**Satri-cel/CT041** *CLDN18.2* GC,PC CARsgen Therapeutics	**AB-2100** *PSMA/CA9* RCC Arsenal Biosciences
QEL-001 *HLA-A2* Liver Transplant Quell Therapeutics	**Zorpocabtagene autoleucel** *CD19* SLE, SSc, PD Miltenyi Biomedicine	**KITE-363** *CD19/CD20* SLE, LN, SSc, IIM Kite, A Gilead Company	**AZD0120** CD19/BCMA MM, amyloid light-chain amyloidosis AstraZeneca	**GLPG5301** *BCMA* MM Galapagos	**A2B543** *MSLN+IL-12* Solid Tumors A2 Biotherapeutics	**Rese-cel** *4-1BB CD19* SLE, SSc, Myasthenia Gravis, Myositis Cabaletta Bio
Rapcabtagene Autoleucel *CD19* MS, MG, RA, SD Novartis	**LEU011** *NKG2D* Solid Tumours Leucid Bio	**miv-cel** *CD19* RA, LN, SSc, ANCA associated vasculitis, Kyverna Therapeutics	**AZD0754** *STEAP2* Prostate cancer AstraZeneca	**JNJ-90014496** *CD19/CD20* B-NHL J&J	**CHM-2101** *CDH17* GC, CRC, Neuroendocrine tumors Chimeric Therapeutics	**AZD0120** *CD19/BCMA* SLE AstraZeneca

Phase 1/2 (n=31)

Note: Assets in development in the US and EU
^ANCA associated vasculitis

Immunology | Hem malignancies | Solid Tumors

Allogeneic
Autologous

Abbreviations: DLBCL: Diffuse Large B-Cell Lymphoma; SLE: Systemic Lupus Erythematosus; SSc: Systemic Sclerosis; LN: Lupus Nephritis; T-ALL: T-cell Acute Lymphoblastic Leukemia; T-LBL: T-cell Lymphoblastic Lymphoma; RRpPCL: r/r primary Plasma Cell Leukemia; IIM: Idiopathic Inflammatory Myopathy; MS: Multiple Sclerosis; MG; Myasthenia Gravis; RA: Rheumatoid Arthritis; SD: Sjogren's Disease; PD: , Polymyositis Dermatomyositis

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What's Ahead in 2026? (1/2)

Asset	Company	Indication	Upcoming Milestones
Deramiocel (CAP-1002)	Capricor Therapeutics	DMD cardiomyopathy	PDUFA target action date is set for Aug 22, 2026 & upon approval, Capricor also eligible to receive priority review voucher
CASGEVY	CRISPR Therapeutics / VERTEX	SCD or TDT (in patients aged 5-11 years)	Global regulatory submissions are anticipated to commence in 1H 2026 by Vertex Pharmaceuticals
Zugocabtagene geleucel (zugo-cel/CTX112)	CRISPR Therapeutics	Autoimmune: SLE, SSc, inflammatory myositis, ITP wAIHA Immuno-oncology: B-cell malignancies	Updates from autoimmune P1 trials of Zugo-cel (SLE, SSc, and inflammatory myositis), P1 ITP trial, and immuno-oncology P1/2 trial in B-cell malignancies expected in 2H 2026
Ebvallo (tab-cel)	ATARA BIO	PTLD	US FDA Regulatory update is expected in 2Q 2026
Orca-T	orcabio	Hematological malignancies: AML, ALL & MDS	Expected PDUFA target action date is Apr 6, 2026
Prulacabtagene leucel (prula-cel/ADI-001)	Adicet Bio	LN, SLE, SSc, IIM, SPS, ANCA-AAV & RA	Clinical updates from P1 are anticipated in 1H 2026 with further clinical updates in 2H 2026
ADI-212	Adicet Bio	mCRPC	Regulatory filing planned for 1H 2026
Prulacabtagene leucel (prula-cel/ADI-001	Adicet Bio	RA	P1 clinical update in 2H 2026
Prulacabtagene leucel (prula-cel/ADI-001)	Adicet Bio	LN, SLE	Expected to engage with US FDA in 2Q 2026 regarding potential pivotal study design, followed by planned initiation of a pivotal trial in 2H 2026
Rese-cel	Cabaletta Bio	SLE, LN & gMG	Complete P1/2 cohort clinical readouts from RESET-SLE, RESET-SSc, and RESET-MG anticipated in 1H 2026
Rese-cel	Cabaletta Bio	PV, SLE & LN	RESET-PV new durability & initial higher-dose clinical data without preconditioning expected in 1H 2026 and RESET-SLE dose-ranging data anticipated in 2026

Abbreviations: DMD: Duchenne Muscular Dystrophy; SCD: Sickle Cell Disease; TDT: Thalassemia; SLE: Systemic Lupus Erythematosus; SSc: Systemic Sclerosis; ITP: Immune Thrombocytopenia Purpura; WAIHA: warm Autoimmune Hemolytic Anemia; PTLD: Post-Transplant Lymphoproliferative Disease; AML: Acute Myeloid Leukemia; ALL: Acute Lymphoblastic Leukemia; MDS: Myelodysplastic Syndromes; LN: Lupus Nephritis; IIM: Idiopathic Inflammatory Myopathy; SPS: Stiff Person Syndrome; ANCA-AAV: Anti-Neutrophil Cytoplasmic Autoantibody Associated Vasculitis; RA: Rheumatoid Arthritis; mCRPC: metastatic Castration-Resistant Prostate Cancer; PV: Pemphigus Vulgaris; BLA: Biologics License Application; GMP: Good Manufacturing Practice

What's Ahead in 2026? (2/2)

Asset	Company	Indication	Upcoming Milestones
Cemacabtagene ansegedleucel	Allogene Therapeutics	LBCL	Interim futility analysis from the pivotal P2 ALPHA3 trial in first-line (1L) is planned for early Apr 2026
ALLO-329	Allogene Therapeutics	SLE, LN, scleroderma & inflammatory myositis	Proof-of-concept data expected in Jun 2026
CTD402	Imviva Biotech	R/R T-ALL/LBL	P1b interim results expected by mid-2026, while the TENACITY-01 study is projected to complete by late 2028
Bria-IMT	BriaCell	MBC	Topline interim results from P3 study expected in 1H 2026
Anito-cel	Arcellx	MM	PDUFA target action date is set for Dec 23, 2026
Mivocabtagene autoleucel, (KYV-101)	kyverna	SPS	BLA submission expected in 1H 2026 & launch by 2026 end, with full SPS data set expected to be presented at a medical conference in 2026
Mivocabtagene autoleucel, (KYV-101)	kyverna	SPS, gMG	Company to present KYSA-8 registrational P2 primary analysis & additional long-term P2 follow-up data from KYSA-6 trial at AAN 2026
Mivocabtagene autoleucel, (KYV-101)	kyverna	progressive multiple sclerosis, RA & LN	Kyverna plans to report additional clinical data updates from P1 IIT, P2 IIT & P1 data In 2026
KYV-102	kyverna	Rapid whole blood process	In 2026, outlining the development strategy for KYV-102
NXC-201	IMMIX Biopharma	R/R AL amyloidosis	Final data from P2 NEXICART-2 trial expected in 2026, followed by a planned BLA submission

Abbreviations : SLE: Systemic Lupus Erythematosus; LN: Lupus Nephritis; gMG: generalized Myasthenia Gravis; LBCL: Large B-cell Lymphoma; T-ALL: T-cell Acute Lymphoblastic Leukemia; LBL: Lymphoblastic Lymphoma; mBC: metastatic Breast Cancer; MM: Multiple Myeloma; SPS: Stiff Person Syndrome; RA: Rheumatoid Arthritis; AL: Amyloid Light-Chain; BLA: Biologics License Application; AAN: American Association of Neurology

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Barriers & Regulatory Shifts

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Designations:

Cell Therapy Advances Accelerate with Major US, EU Designations in 2025

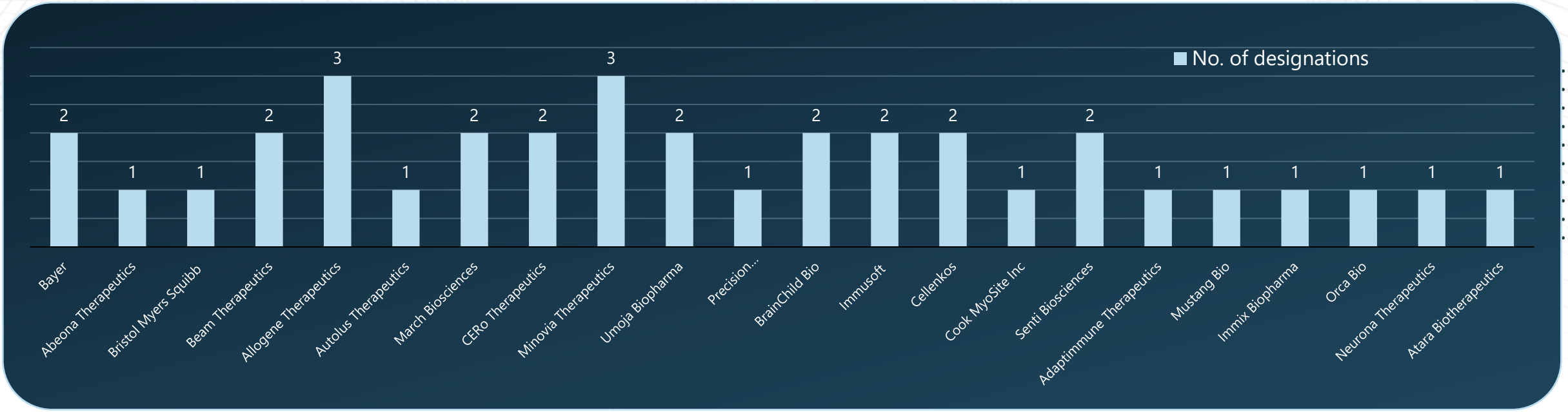
Key Takeaways

- Majority of designations are from the US FDA highlighting strong reliance on expedited pathways (FTD, RMAT) to accelerate cell therapy development in high unmet-need areas
- A large proportion of therapies target hematologic malignancies such as AML, lymphoma and leukemia (eg:- programs from Senti Biosciences, Umoja Biopharma)
- Cell therapies are expanding beyond oncology, with companies like Immusoft and Neurona Therapeutics driving growth across neurology, rare diseases & other therapeutic areas

FTD	ODD	Priority Review
11	9	3

BTD	RMAT	RPDD/PRIME
2	6	3

Company-wise Distribution



Abbreviations: FTD Fast Track Designation | ODD – Orphan Drug Designation | BTD – Breakthrough Designation | RMAT - Regenerative Medicine Advanced Therapy Designation, PRIME – Priority Medicine Designation | RPDD - Rare Pediatric Disease Designation

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What's New in CGT Regulation?

Key updates from US FDA, EMA & ICH shaping the future of cell & gene therapy

U.S. FOOD & DRUG ADMINISTRATION

- **Jan, 2026: From Lab to Market Faster: Flexible CMC Framework for CGT:** US FDA is using a more flexible approach to CMC requirements for CGT, like clinical development, commercial specifications & process validation flexibilities, to help companies better prepare for BLA submission & approval
- **Sep, 2025: Fast-Track Regulatory Programs for Regenerative Medicine:** This guidance helps developers of regenerative medicine therapies for serious or life-threatening conditions understand the US FDA expedited programs, including RMAT designation. It explains how to use accelerated approval pathways, provides advice on clinical development & highlights opportunities to engage with CBER to speed up the development & review process
- **Sep, 2025: Beyond Approval: Capturing Real-World Data for CGT Products:** The aim is to explain methods for collecting safety and efficacy data for CGT products after approval, including LFTU (e.g., 15 years). (This guidance does not cover data collected to support the expansion of clinical indications)
- **Sep, 2025: Innovative Trial Designs for CGT in Small & Rare Populations:** US FDA provide recommendations to sponsors planning clinical trials for CGT products used in small or rare patient populations. It explains FDA requirements and outlines different trial designs (Adaptive/Bayesian trial design & Master protocols) and surrogate endpoints that can be used to generate the clinical evidence needed for product approval. (This document elaborates on the US FDA previously established guidance principles)

ICH harmonisation for better health

- **Dec, 2025: Future Guideline Roadmap for ATMPs:** ICH CGTDG, first established key guiding principles for ATMPs & then created a roadmap for CGT, focusing on priority areas that need better harmonization along with recommendations for new or updated guidelines
- **May, 2025: ATMP Comparability: Proposed New ICH Q5E Annex:** ICH Multidisciplinary guideline (ICH Q5E) is proposed to address the unique development & regulatory challenges of ATMPs, including CGTs, with a focus on ensuring comparability when their manufacturing processes changes

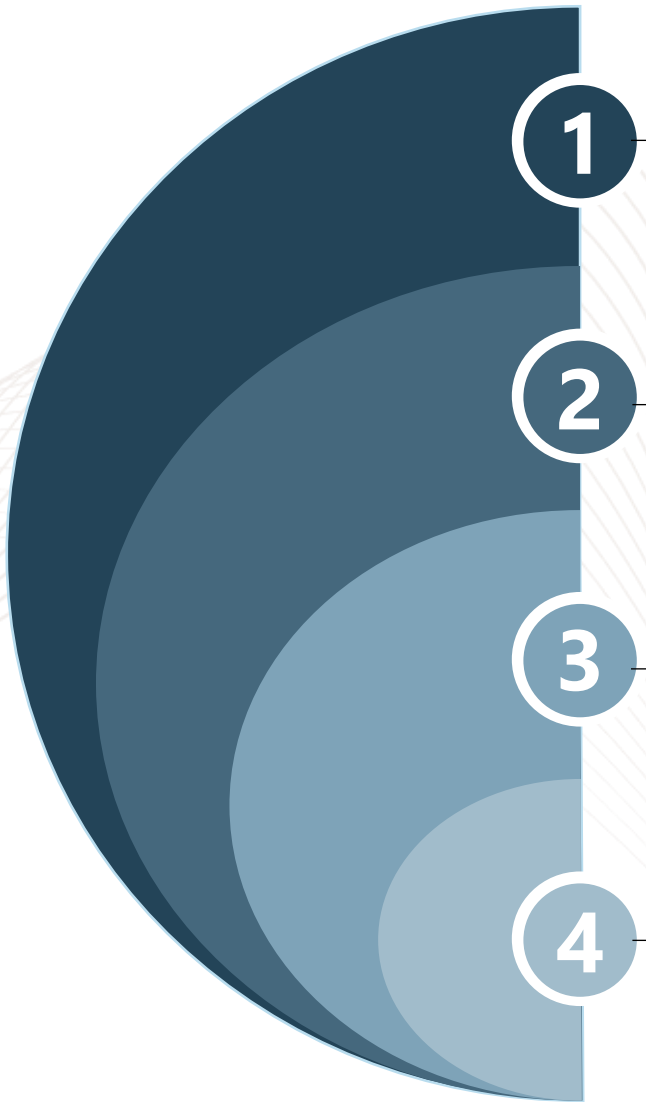
EUROPEAN MEDICINES AGENCY SCIENCE MEDICINES HEALTH

- **Jan, 2025: EMA Sets Standards for ATMP Clinical Trials (Effective July 1, 2025):** EMA has introduced a multidisciplinary guideline outlining the development, manufacturing, quality control, non-clinical & clinical requirements for investigational ATMPs. It explains what data and structure are needed for a CTA (both early and late-stage studies) helping companies prepare for a MAA
- **Mar, 2025: EMA Revises GMP Framework for ATMPs:** EMA has released a concept paper proposing updates to Part IV of the EU GMP guidelines for ATMPs, aiming to align with current Annex 1, while maintaining flexibility in ATMP production

Abbreviations: CMC: Chemistry, Manufacturing and Control; CGT: Cell and Gene Therapies; BLA: Biologics License Application; RMAT: Regenerative Medicine Advanced Therapy; CBER: Center for Biologics Evaluation and Research; LFTU: Long-Term Follow-Up; ATMPs: Advanced Therapy Medicinal Products; CTA: Clinical Trial Application; MAA: Marketing Authorization Applications; GMP: Good Manufacturing Practice; CGTDG: Cell and Gene Therapy Discussion Group

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Key Barriers Slowing Cell Therapy Trials



1. **Biological and Immunological Barriers**

   Cell therapies often fail because the patient's body environment can suppress or block therapeutic cells, reducing their ability to expand and function *in vivo*. Host-related factors such as immune regulation, prior treatments, and variable patient biology also contribute to reduced efficacy or relapse

2. **Target Antigen Instability and Disease Heterogeneity**

   Many programs fail due to antigen escape, where disease cells alter or lose target antigens. Additionally, heterogeneous antigen expression makes it difficult to identify targets specific to diseased cells without affecting healthy tissue

3. **Limited *In Vivo* Persistence, Trafficking, and Functional Durability**

   Even when cell products are successfully infused, they may not reach the right tissues in sufficient numbers, fail to persist, or become functionally exhausted over time. Inadequate product fitness (for example, poor-quality cells at the start or cells not in an ideal state) and continuous antigen stimulation can drive T-cell exhaustion and loss of anti-disease activity

4. **Safety, Manufacturing, and Regulatory Constraints**

   Trials frequently stall or fail due to toxicity (such as cytokine release syndrome and neurotoxicity), inconsistent manufacturing, or inability to meet evolving regulatory standards for advanced therapies. Challenges in production, supply chain, and quality validation can delay or halt progress despite early efficacy signals

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Cell Therapy:
Hematological Malignancies Vs Solid Tumor

Antigen Heterogeneity:



Solid tumors are highly heterogeneous, meaning cancer cells within the same tumor or between tumors in one patient can express different levels of target antigens.
This variability makes it difficult to identify a single tumor-associated antigen for CAR-T therapy, often leading to antigen escape, incomplete tumor killing, and potential relapse.

Cell Trafficking & Physical Barriers



CAR-T cells need to effectively reach, infiltrate, and persist at tumor sites to function. In blood cancers, they circulate freely and easily interact with tumor cells. However, solid tumors create physical barriers like dense stroma and abnormal vasculature that limit cell entry. Tumor architecture further restricts deep penetration into the tumor core. As a result, CAR-T cell activity is significantly reduced in solid tumors.

Safety concerns



CAR-T therapies in hematological malignancies commonly cause CRS and ICANS, which are usually manageable with proper treatment. In contrast, solid tumors carry higher risks of on-target, off-tumor toxicity due to shared antigen expression with normal tissues. This can damage vital organs, making safety a major challenge in solid tumor settings.

Abbreviations: CRS: Cytokine Release Syndrome, ICANS: Immune Effector Cell-Associated Neurotoxicity Syndrome

Tumor Microenvironment



The TME suppresses immune activity through inhibitory cells and cytokines (e.g., TGF-β, IL-10), limiting T-cell function. In solid tumors, a dense and structured environment restricts CAR-T infiltration and activity. In contrast, hematological TMEs are more diffuse and dynamic enabling continuous cell interactions but also supporting immune evasion and resistance.

T-cell Exhaustion



T-cell exhaustion is more pronounced in solid tumors due to continuous antigen exposure and an immunosuppressive microenvironment. This leads to increased inhibitory receptors (e.g., PD-1, CTLA-4, TIM-3) and reduced effector function. As a result, CAR-T cells show diminished cytotoxic activity and persistence. In contrast, lower exhaustion in hematological malignancies allows sustained activity and improved therapeutic efficacy.

Cell persistence and optimisation



In hematological malignancies, CAR-T cells show strong expansion and long-term persistence, especially with early-memory T-cell enrichment. In solid tumors, persistence is limited due to T-cell exhaustion and TME-driven dysfunction. To address this, next-generation approaches include armored CAR-T cells and dominant-negative TGF-β receptors. These strategies aim to enhance durability and are showing early clinical promise.

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SWOT Analysis: Cell Therapy

S

Strengths

- **Bridging the gap in unmet medical needs:** Offer vital hope for patients with rare disorders and hard-to-treat cancers lacking other options, meeting key unmet needs
- **Expanding pipelines beyond hematology:** Numerous clinical trials are actively exploring cell therapies in autoimmune diseases and immunology
- **Single-Treatment Cure Potential:** Certain once-incurable diseases are now curable
- **Alignment with personalized medicine:** Autologous therapies customized for individual patients

W

Weakness

- **High cost:** With therapies costing hundreds of thousands to millions $ per dose, strain healthcare systems and limit equitable access
- **Complex manufacturing and supply chain:** Cell therapies as "living" drugs, demand complex cold-chain logistics, manual manufacturing prone to costly failures, uncertain biological integration, and intricate supply chains for quality and efficacy
- **Side effects:** Triggers immune side effects and severe reactions like CRS or ICANS, while allogeneic therapies risk graft-versus-host disease

O

Opportunity

- **Expansion into solid tumor:** After a decade modest progress in solid tumor, cell therapies are now showing promise in tough cases like glioblastoma, with TILs and TCR-T technologies gaining potential
- **AI-driven personalization revolution:** accelerating therapy development, enhancing precision and safety, and reducing the time required for clinical trials
- **Innovative Payment Models:** Payers test outcomes-based payments and installments to manage high-cost, boosting access and cutting risks
- **Allogeneic "Off-the-Shelf" revolution:** Enable scalable, uniform manufacturing that lowers costs and speeds up patient treatment

T

Threats

- **Regulatory and reimbursement uncertainty:** Evolving standards, regional differences, and payer pushback on pricing create commercial and development risk
- **Physician & Patient Adoption:** Doctors skip risky cell therapies for long-term illnesses if safer treatments work; patients avoid them due to high costs, long hospital time, and prep steps when cheaper options exist
- **Competition from other modalities:** Competition from ADCs and BiTE antibodies, which enable straightforward drug administration

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Value & Commercialization

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Market Pricing & Reimbursement

US payers use diverse, innovative payment models for cell and gene therapies to manage high upfront costs, share financial risk, and preserve patient access. Innovative approaches like outcomes-based agreements link reimbursement to clinical success, enhancing affordability and accountability.

Market Context

- US access is shaped by a multi-payer system like Medicare, Medicaid and commercial insurers, each use different coverage and payment pathways
- For FY 2025, inpatient cases involving CAR-T therapy are classified under MS-DRG 018, with a base reimbursement amount of $269,139
- The base payment will increase by 17% to $314,231 in FY 2026, up from $269,139 in FY 2025 for inpatient CAR-T treatment

Price Level/Cost Burden

- Research findings in 2024 shows median overall costs above $600,000, with 8.5% of patients exceeding $1M. CAR-T therapies such as KYMRIAH and YESCARTA are priced at $373,000 to $475,000 per infusion, while ABECMA and CARVYKTI exceed $450,000 to $465,000, with total patient costs often surpassing $1M after additional hospitalization and management cost in 2026
- In 2024, under Medicare, CAR-T therapy claims averaged $498,723 in inpatient settings ranging from $276,138 to $1,066,524, compared to $414,393 in outpatient settings ranging from $276,980 to $849,878

Implemented Models	Pros	Cons
Outcome Based Agreement (OBA)/Value Based Contracts	• Payers pay only if outcomes are achieved; lowers risk of ineffective high-cost therapy	• Difficult to agree on measurable, clinically meaningful endpoints and timelines • Patient follow-up is hard (e.g., plan switching affects data continuity)
Warranty Model	• Reimburses costs for suboptimal outcomes if the treatment does not work as expected	• Do not address upfront affordability • Requires tracking and identifying non-responding patients
Subscription Model	• Fixed subscription fee provides stable and manageable budget impact • Spreads financial risk across populations and multiple payers	• Payers may incur extra costs regardless of actual usage • May shift risk unnecessarily if payer can already manage it internally
Payment Over Time/Installment Model	• Spreads high upfront costs into smaller payments, making it easier for smaller payers to manage budgets	• Less favored by larger payers, who prefer upfront payment within a single budget cycle, and it is not linked to patient outcomes
Cell and Gene Therapy (CGT) Access Model	• CMS-led negotiation, funding, and technical support for states • Helps address outcomes tracking and reporting challenges	• Limited by scope as currently focused mainly on sickle cell disease therapies

Abbreviations: MS-DRGs: Medicare Severity Diagnosis Related

Note: Some models are still in pilot or partially implemented stages but are grouped under implemented models for simplicity

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Market Pricing & Reimbursement: Key Access Barriers

High cost & inadequate reimbursement: High costs and inadequate MS-DRG reimbursements create affordability challenges for providers and payers, underscoring the need for better cost clarity and planning



Payment delays & cash flow issues: Delayed reimbursement timelines strain provider cash flow, creating financial risk and potentially delaying patient treatment



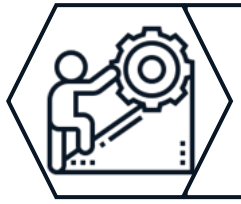
Medicaid variability: Medicaid coverage, management, reimbursement and financing variability between federal and states



Complex negotiations: Complex payer–provider negotiations create reimbursement uncertainty and risk of financial losses, especially when treatment costs vary by patient needs



Coverage challenges: Limits of stop-loss coverage and payer pushback on 340B (drug pricing program) discounts restrict access and create reimbursement uncertainty



Insufficient Data: A persistent challenge is the lack of long-term real-world data to justify the high upfront costs



Abbreviations: MS-DRG - Medicare Severity Diagnosis-Related Group | OBA – Outcome-Based Agreement

Value, Outcomes and RWE in CAR-T



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Journal homepage: www.elsevier.com/locate/jval

Systematic Literature Review

Economic Evaluations of Chimeric Antigen Receptor T-Cell Therapies for Hematologic and Solid Malignancies: A Systematic Review

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ABSTRACT

Objectives: This study aimed to systematically review evidence on the cost-effectiveness of chimeric antigen receptor T-cell (CAR-T) therapies for patients with cancer.

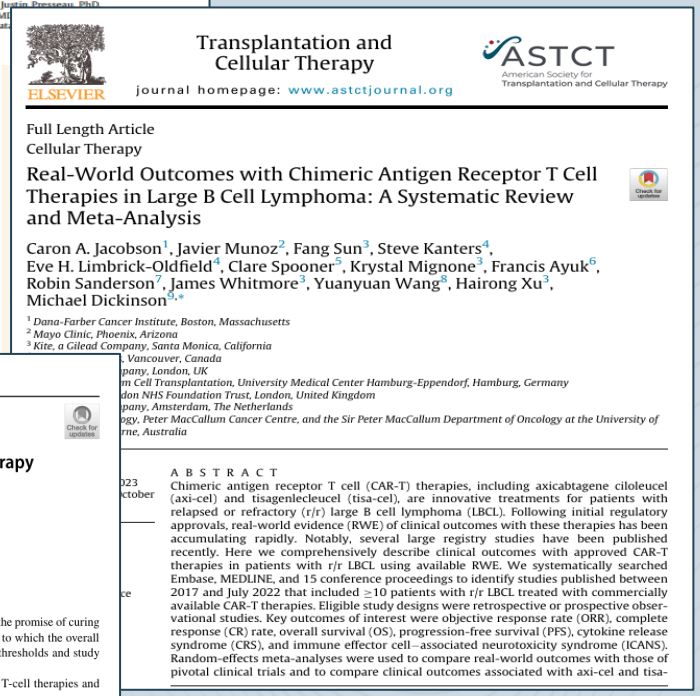
Methods: Electronic databases were searched in October 2022 and updated in September 2023. Systematic reviews, health technology assessments, and economic evaluations that compared costs and effects of CAR-T therapy in patients with cancer were included. Two reviewers independently screened studies, extracted data, synthesized results, and critically appraised studies using the Philips checklist. Cost data were presented in 2022 US dollars.

Results: Our search yielded 1809 records, 47 of which were included. Most of included studies were cost-utility analysis, published between 2018 and 2023, and conducted in the United States. Tisagenlecleucel, axicabtagene ciloleucel, idecabtagene vicleucel, ciltacabtagene autoleucel, lisocabtagene maraleucel, brexucabtagene autoleucel, and relmacabtagene autoleucel were compared with various standard of care chemotherapies. The incremental cost-effectiveness ratio (ICER) for CAR-T therapies ranged from $9424 to $4 124 105 per quality-adjusted life-year (QALY) in adults and from $20 784 to $243 177 per QALY in pediatric patients. Incremental cost-effectiveness ratios were found to improve over longer time horizons or when an earlier cure point was assumed. Most studies failed to meet the Philips checklist due to a lack of head-to-head comparisons and uncertainty surrounding CAR-T costs and curative effects.

Conclusions: CAR-T therapies were more expensive and generated more QALYs than comparators, but their cost-effectiveness was uncertain and dependent on patient population, cancer type, and model assumptions. This highlights the need for more nuanced economic evaluations and continued research to better understand the value of CAR-T therapies in diverse patient populations.

Keywords: cancer, chimeric antigen receptor T-cell therapies, economic evaluation, systematic review.

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Transplantation and Cellular Therapy

journal homepage: www.astctjournal.org

ASTCT American Society for Transplantation and Cellular Therapy

Full Length Article
Cellular Therapy

Real-World Outcomes with Chimeric Antigen Receptor T Cell Therapies in Large B Cell Lymphoma: A Systematic Review and Meta-Analysis

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ABSTRACT

Chimeric antigen receptor T cell (CAR-T) therapies, including axicabtagene ciloleucel (axi-cel) and tisagenlecleucel (tisa-cel), are innovative treatments for patients with relapsed or refractory (r/r) large B cell lymphoma (LBCL). Following initial regulatory approvals, real-world evidence (RWE) of clinical outcomes with these therapies has been accumulating rapidly. Notably, several large registry studies have been published recently. Here we comprehensively describe clinical outcomes with approved CAR-T therapies in patients with r/r LBCL using available RWE. We systematically searched Embase, MEDLINE, and 15 conference proceedings to identify studies published between 2017 and July 2022 that included ≥10 patients with r/r LBCL treated with commercially available CAR-T therapies. Eligible study designs were retrospective or prospective observational studies. Key outcomes of interest were objective response rate (ORR), complete response (CR) rate, overall survival (OS), progression-free survival (PFS), cytokine release syndrome (CRS), and immune effector cell–associated neurotoxicity syndrome (ICANS). Random-effects meta-analyses were used to compare real-world outcomes with those of pivotal clinical trials and to compare clinical outcomes associated with axi-cel and tisa-



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https://doi.org/10.1007/s41669-025-00614-x

SYSTEMATIC REVIEW

Cost-Effectiveness of Chimeric Antigen Receptor (CAR) T-Cell Therapy for Blood Cancers: An Updated Systematic Review

Nishma Patel[1] · Suzanne Farid[2] · Manuel Gomes[1]

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Abstract

Background Chimeric antigen receptor (CAR) T-cell therapy is an area of rapid development, showing the promise of curing blood cancers. While substantial health gains may justify high costs, it is currently unclear the extent to which the overall cost effectiveness of these therapies is driven by i) context-specific factors, such willingness-to-pay thresholds and study perspective, or ii) important subgroups such as line of treatment and therapy product.

Objective This paper aims to critically review published evidence on the cost effectiveness of CAR T-cell therapies and assess the key factors that drive their cost effectiveness.

Methods We conducted a systematic review using PubMed, Scopus and Ovid (Embase) databases to identify full economic evaluations of CAR T-cell therapies published up to January 2024. One reviewer screened and extracted data from the studies and the second reviewer assessed a sample of the full-text studies against the inclusion/exclusion criteria. Studies were critically appraised using the CHEERS checklist. Cost data are presented in 2022 US dollars.

Results The review identified 45 full cost-effectiveness studies of CAR T-cell therapies. These studies considered a total of 92 treatment comparisons, which included tisagenlecleucel ($n = 37$), axicabtagene ciloleucel ($n = 28$), brexucabtagene autoleucel ($n = 7$), lisocabtagene maraleucel ($n = 8$), idecabtagene vicleucel ($n = 6$), ciltacabtagene autoleucel ($n = 4$) and relmacabtagene autoleucel ($n = 2$). Incremental cost ranged from − US$74,980 to US$714,178 and incremental quality-adjusted life year (QALY) gains ranged from − 0.02 to 10.77. The resulting cost-per-QALY-gained ratios ranged from − US$37,490,000 to US$7,972,845, and the range of willingness-to-pay (WTP) thresholds between US$36,184 to US$317,825. The price of CAR T-cell therapy represented 75% (mean US$391,060) of the total cost of CAR T-cell therapy but was not the sole factor influencing cost effectiveness. Hospitalisation made up 6% of the total cost (mean US$34,152), while adverse events accounted for 9% (mean US$47,350). Regression analysis indicated cost effectiveness did not change according to important clinical or contextual factors.

Conclusions The findings demonstrate that the cost effectiveness of CAR T-cell therapies is determined by a combination of factors: the relative difference between the cost of the CAR T-cell therapy and comparator, the magnitude of the QALY gains and the WTP thresholds. Their cost- effectiveness does not differ according to therapy product, line of treatment, or country.

Abbreviations: QALY: Quality-Adjusted Life Year; PFS: Progression-Free Survival; ICANS: Immune Effector Cell-Associated Neurotoxicity Syndrome; ICERs: Incremental Cost-Effectiveness Ratio

Key Findings:

Value driven by core economic levers: Cost-effectiveness of CAR-T therapies hinges primarily on QALY gains, incremental costs vs comparators, and country-specific willingness-to-pay thresholds and not on product type, indication, or geography

High upfront costs dominated by drug price: Drug acquisition accounts for ~75% of total cost (~$391k), while hospitalization (~6%) and adverse event management (~9%) are smaller but still meaningful contributors to overall cost-effectiveness

Strong real-world effectiveness: Meta-analyses show CD19 CAR-T therapies maintain robust efficacy and manageable safety in broader, real-world populations beyond clinical trials

Efficacy–safety trade-offs between products: Axi-cel demonstrates superior efficacy (e.g., ~40% lower mortality risk, better PFS and response rates) compared to tisa-cel, but with significantly higher rates of severe neurotoxicity (ICANS)

Higher cost, higher benefit vs standard care: Across 47 studies covering 7 CAR-T therapies, these treatments consistently deliver greater QALYs but at substantially higher costs than conventional therapies

Wide ICER variability and uncertainty: ICERs range widely (≈$9k to >$4M/QALY in adults), reflecting sensitivity to assumptions, patient populations, and cancer types - underscoring persistent uncertainty and the need for refined economic modeling

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Funding Dashboard 2025

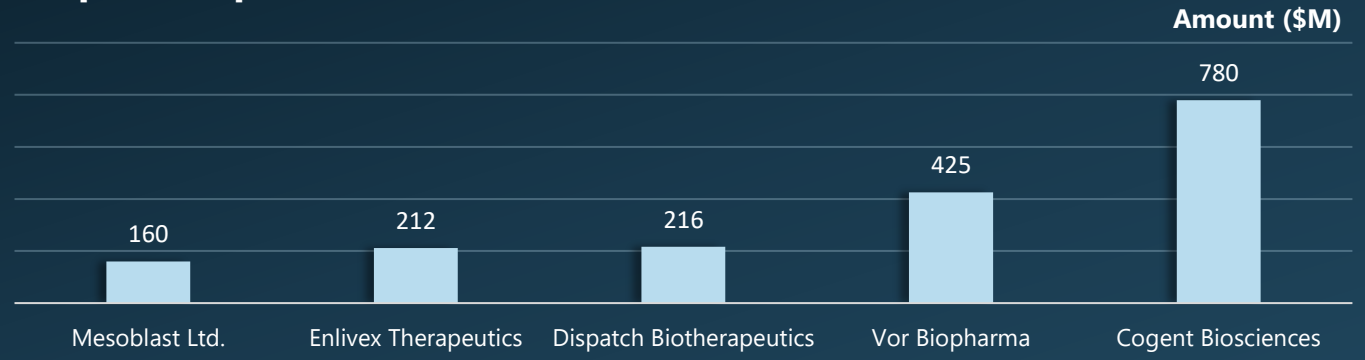
Key Highlights

- Most financings are clustered in P2, followed by a sizeable number of Preclinical/IND and P1 programs
- RA Capital stands out as the most active repeat investor, funding early rounds for Azalea Therapeutics (Seed/Series A) and late-stage rounds for multiple companies (eg:- Be Biopharma, Vor Biopharma, Gameto)
- Majority of investment is toward later-stage financings like Common and PIPE, together with larger Series B/C and Series Unspecified/Private rounds
- Cogent Biosciences is heavily financed as a late-stage player, securing multiple large P3-stage raises (Debt/Loan, Common, Convertible Notes) in 2025

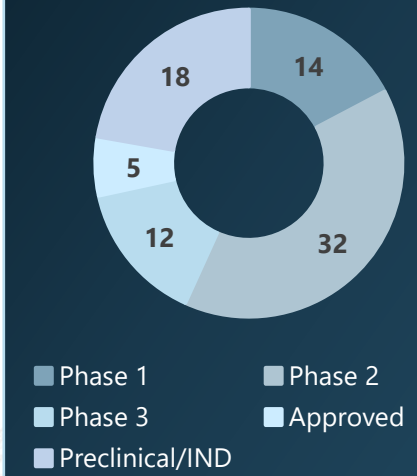
Total Amount
$~5.2B

No. of Companies
87

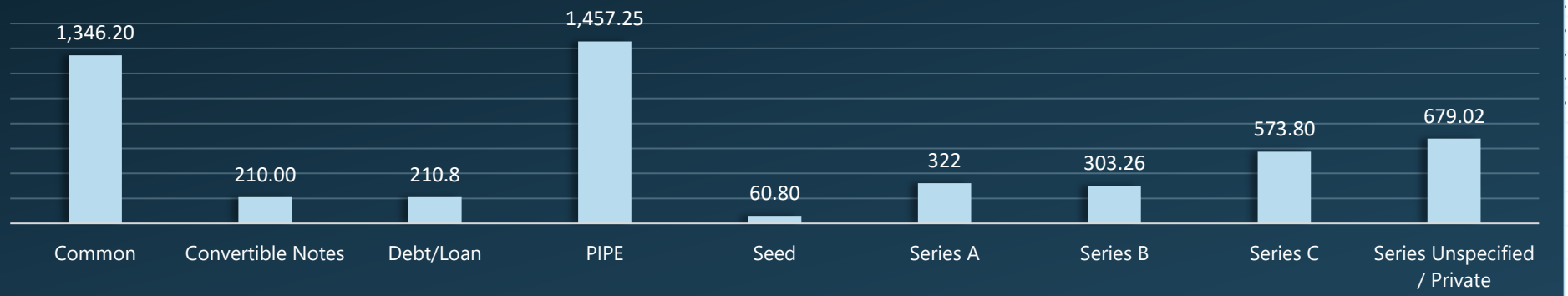
Top 5 Companies



Phase-wise (Companies)



Round-wise Amount ($M)



Note: 1. Company counted only once for no. of companies & for phases section; 2. Total amount include all the amount that companies received once or repeatedly in 2025; 3. Round-wise section include all the amount that companies received once or repeatedly in that round in 2025; 4. Top 5 companies means companies that received most funding once or repeatedly in 2025; 5. Cell therapy and cell & gene therapy included for this funding data

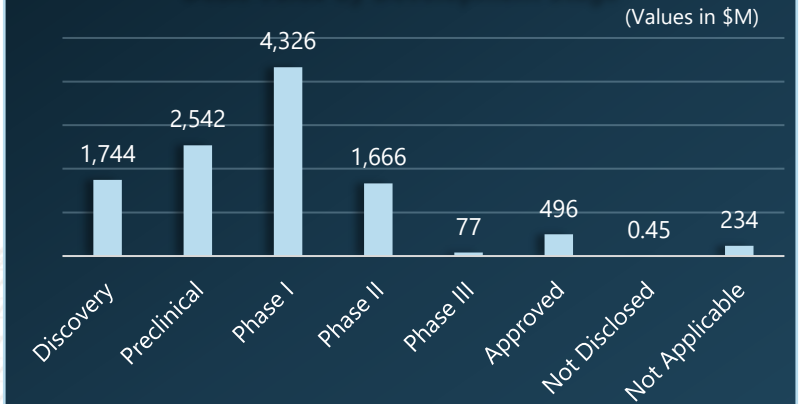
This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

INSIGHTS CORNER

Deal Dashboard 2025

204 Total Deals Across all Stages	$ 11.08 B Total Deal Value	$ 5.45 B Total Upfront	33 Rare Diseases Deals (15% of Total deals)	105 Contract Services (Top Deal Type)	104 Oncology Deals (Top Therapy Area)

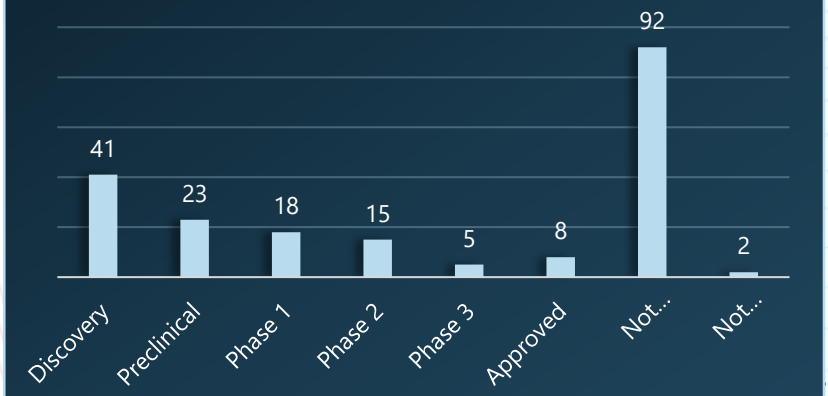
Deals Value by Development Stage



Not applicable includes contract service, university, business unit purchase, platform product purchase

Deals Type Breakdown	No. of Deals
Contract Services	105
Dev and Commercial Licensing	28
University/Foundation	28
Acquisition	16
R&D Only	12
Joint Venture	4
Testing and CT Agreement	3
Company Format	2
Reverse Merger	2
Platform and Product Purchase	1
Business Unit Purchase	1
Royality Stream Purchase	1

No. of Deals by Development Stage

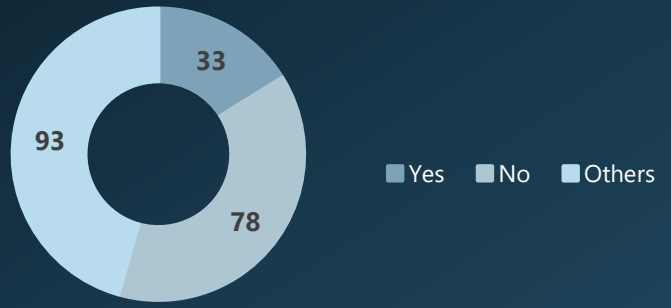


Global Deal Value by Region



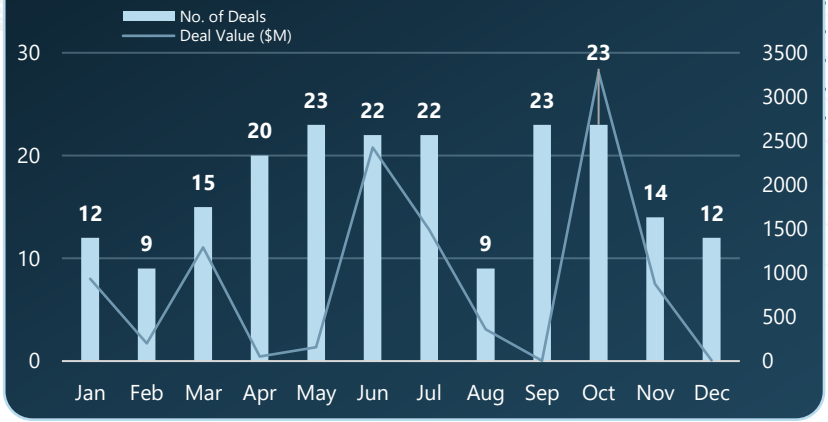
Undisclosed Deals $3200 M (Region was not specified)

Rare Diseases Deals



Others include Unknown (40), Not Applicable (53)

Trend Analysis of Deal Count and Deal Value



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This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com



Ecosystem, Company Journey, & Outlook

INSIGHTS CORNER

Evolution story: Kite Pharma

Headquarter
Santa Monica, CA

Kite Pharma's evolution is essentially the story of one of the first dedicated CAR-T cell-therapy companies, from a small immunotherapy startup to a Gilead-owned global leader in autologous cell therapy

Kite
A GILEAD Company

2009

Founded in Santa Monica, CA by Arie Belldegrun

2012

Kite Pharma signed a research/license deal with NCI for early CAR-T and TCR technologies

2013

Positive earlier results of KTE-C19 P1/2a study for NHL presented at 55th ASH

2014

- Granted ODD to KTE-C19 from US & EU for DLBCL
- Updated results of KTE-C19 P1/2a study for NHL presented at ASCO, which showed 92% ORR

2015

- Kite entered into the strategic research collaboration and license agreement with Amgen to advance next-gen CAR-T cell immunotherapies via its eACT platform for development and commercialization
- Granted ODD to KTE-C19 from EU for PMBL and FL
- Initiated ZUMA-1 P2 study of KTE-C19 for DLBCL, PMBCL and TFL
- Initiated ZUMA-3 P1/2 study of KTE-C19 for r/r ALL
- Granted breakthrough therapy designation from US FDA to KTE-C19 for NHL, DLBCL, PMBCL and TFL

2016

- Kite opened commercial-scale CAR-T facility in El Segundo, CA
- Granted ODD to KTE-C19 from US for PMBCL and FL
- Initiated ZUMA-3 P1/2 study of KTE-X19 for r/r ALL

2017

- Entered into the strategic partnership with Daiichi Sankyo to develop and commercialize KTE-C19 in Japan
- Completed its merger with Gilead and became a wholly-owned subsidiary of Gilead
- Presented clinical data of KTE-C19 in ASCO and ASH
- YESCARTA (KTE-C19) received US FDA approval for DLBCL, PMBCL, and TFL, marking it as the first approved CAR-T therapy in DLBCL

2018-2019

- YESCARTA (KTE-C19) received EU approval for DLBCL and PMBCL
- Presented clinical results of KTE-X19 at ASCO and ASH
- Submitted a BLA to the US FDA for the KTE-X19, for r/r MCL

2020

- Granted Priority Review designation to KTE-X19 for MCL
- TECARTUS (KTE-X19) received US FDA approval for MCL
- TECARTUS (KTE-X19) received conditional approval from EU for MCL

2021-2023

- YESCARTA (KTE-C19) received China approval for DLBCL and PMBCL
- YESCARTA and TECARTUS received approvals in other heme indications from US and EU
- Entered into multiple deals with pharma companies to advance CAR-T and cell therapy pipelines

2024-2026

- Kite remains a Gilead-owned subsidiary and a leading global CAR-T and cell-therapy developer, with YESCARTA and TECARTUS as commercial cornerstones

Ongoing

Asset	Indication	Status
Anito-cel	4L r/r multiple myeloma	BLA
KITE-363	r/r DLBCL	P1
KITE-753	r/r DLBCL	P1
KITE-197	r/r DLBCL	P1

Abbreviations: r/r: Relapsed/Refractory; DLBCL: Diffuse Large B-Cell Lymphoma; NHL: Non-Hodgkin Lymphoma; PMBL: Primary Mediastinal Large B-Cell Lymphoma; TFL: Transformed Follicular Lymphoma; ALL: Acute Lymphoblastic Leukemia; MCL: Mantle Cell Lymphoma; ODD: Orphan Drug Designation; NCI: National Cancer Institute

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

CARVYKTI: Development to Market

CARVYKTI® (ciltacabtagene autoleucel) Suspension for IV Infusion

Johnson&Johnson

BCMA-directed CAR-T

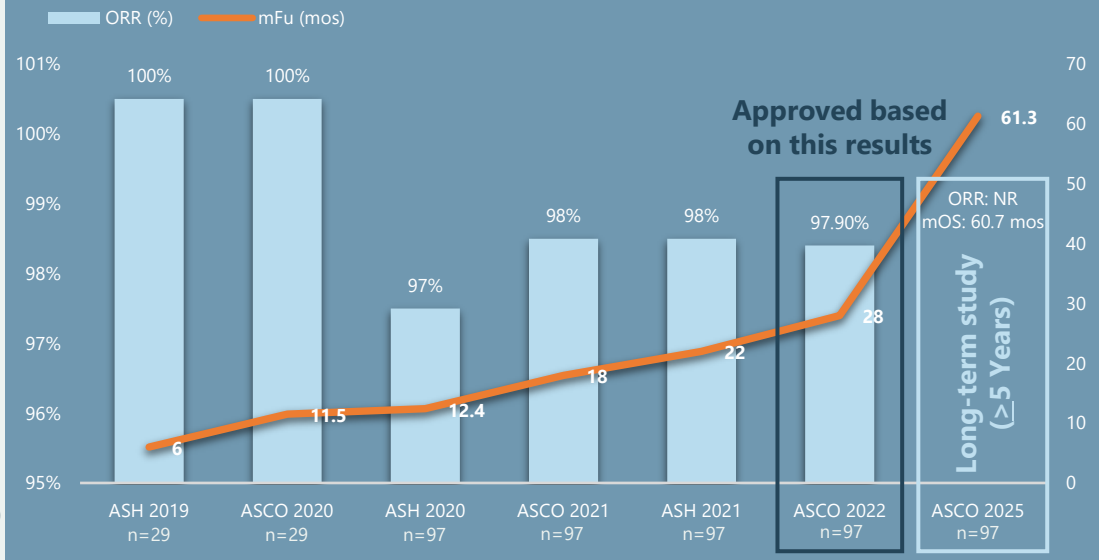
Intravenous

Cell Therapy

Approval Timeline

- 2024 – Approved for RRMM (3L+)
- Approved for RRMM (1L+) – 2024
- 2024 – Approved for RRMM (1L+)
- Approved for RRMM (2L-4L) – 2024
- 2022 – Approved for RRMM (3L+)
- Approved for RRMM (3L+) – 2022
- 2022 – Approved for RRMM (4L+)

Efficacy Success Timeline



	@mos	OS Rate (%)	PFS Rate (%)
ASH 2020	12	89	77
ASCO 2021	18	81	66
ASH 2021	24	74	61
ASCO 2022	27	70.4	54.9

- ORR remains extremely high over time, with early phase 1b data at 91-100% and the pivotal 97-patient cohort sustaining about 97-98% across multiple analyses
- Depth of response deepens with follow-up, as sCR rates rise from roughly mid-60-80% initially to above 80% later, indicating ongoing conversion from partial responses to complete responses
- As data mature, PFS declines from about 77% at 12 mos to the mid-60% range at 18 mos and just over 50% by around 27 mos, whereas OS stays high at roughly 70-80% at 24 mos despite the heavily pretreated population

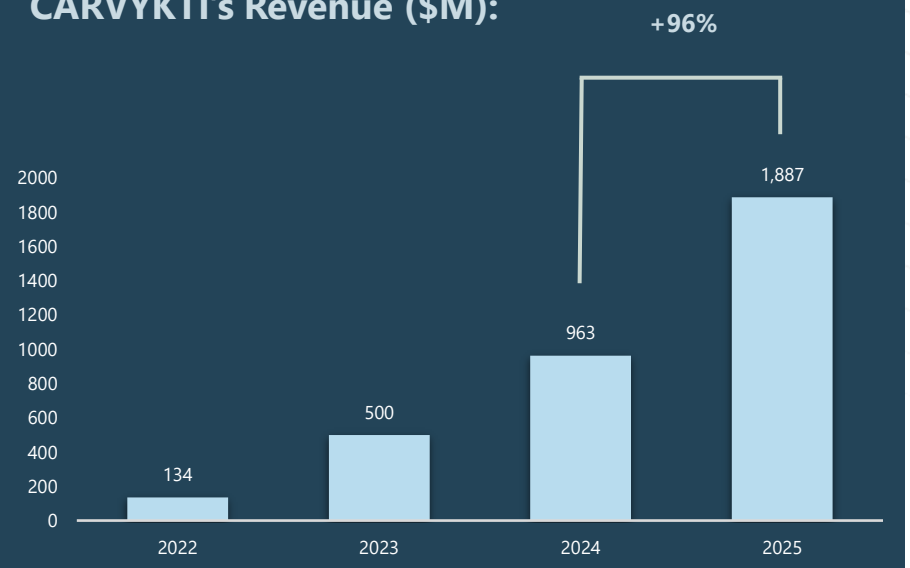
CARVYKTI's Designation:

ODD	2019	BTD/ PRIME	2019
	2020		2019
	2020		2020

In 2025, CARVYKTI's revenue benefits from first full-year contributions of new market launches (e.g., Europe/Asia) alongside broader reimbursement in key markets, while label expansion into earlier lines of therapy and wider patient segments substantially enlarges the eligible population and drives higher uptake.

Abbreviations: mFu: Median follow-up; mos: months; ORR: Overall Response Rate; OS: Overall survival; PFS: Progression free survival; RRMM: Relapsed/Refractory Multiple Myeloma

CARVYKTI's Revenue ($M):



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CARVYKTI Patient Support Resources

CARVYKTI® (ciltacabtagene autoleucel) Suspension for IV Infusion

MYCARVYKTI Patient Support Program	MyCARVYKTI helps eligible US patients and care partners with 24/7 travel-related support during treatment at Activated Treatment Centers, including transportation, lodging, meals, and out-of-pocket expenses for key steps such as leukapheresis, infusion, and bridging therapy. Eligibility requires a CARVYKTI prescription and meeting distance and income criteria, regardless of insurance coverage
CARVYKTI Patient Milestone Guidebook	This guidebook supports patients throughout the treatment journey with key treatment information and helpful resources. It helps patients track progress, capture thoughts & feelings, record appointments & care team contacts, monitor symptoms, & share updates with their healthcare team
CARVYKTI Patient Wallet Card	This card contains important safety information for patients receiving CARVYKTI. Patients should contact their healthcare professional or seek emergency medical help immediately if they experience any possible side effects
Care Partners	Care partners play an important role in supporting patients throughout the CARVYKTI journey. Their support may include driving patients to appointments, especially as driving is restricted for 2 weeks post-infusion, taking notes during doctor visits, and providing emotional encouragement through daily treatment challenges. As patient needs evolve, care partners help improve both treatment adherence and overall experience
CARVYKTI Reimbursement Overview	For its US FDA-approved indication, CARVYKTI is typically covered under Medicare Part A (inpatient) and Medicare Part B (outpatient), with separate reimbursement for the drug and administration. Coverage may also occur through Medicare Part C, depending on prior authorization and site-of-care, with payments following MS-DRG, OPPS, or Physician Fee Schedule rules
Online Portal & Live Assistance for Activated Centers	At activated CARVYKTI treatment centers, a CTEL supports workflow and CQUENCE training, while a CQUENCE Order Specialist manages order tracking. The CQUENCE portal lets authorized centers track the CARVYKTI journey from patient enrollment to infusion

Abbreviations: r/r: Relapsed/Refractory; DLBCL: Diffuse Large B-Cell Lymphoma; NHL: Non-Hodgkin Lymphoma; PMBL: Primary Mediastinal Large B-Cell Lymphoma; CMS: Centers For Medicare & Medicaid Services; CAR-T: Chimeric Antigen Receptor-t Cell; MS-DRG: Medicare Severity Diagnosis Related Group; OPPS: Outpatient Prospective Payment System; PFS: Physician Fee Schedule; CTEL: Cell Therapy Experience Lead

KOL DASHBOARD



Jae Park,

MD, Leukemia Specialist & Cellular Therapist

Education

MD - Johns Hopkins School of Medicine

Affiliation

Memorial Sloan Kettering Cancer Center

Division

Hematology / Oncology

Speciality

Clinical trials | Cellular therapy | Immunotherapy

Email

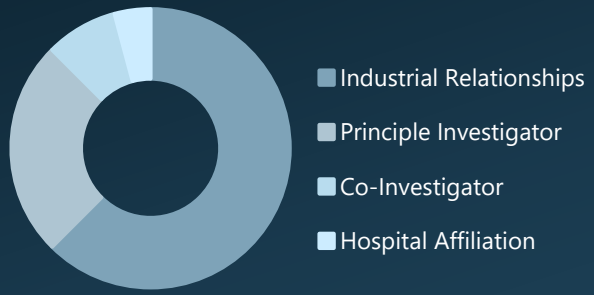
parkj6@mskcc.org

Location

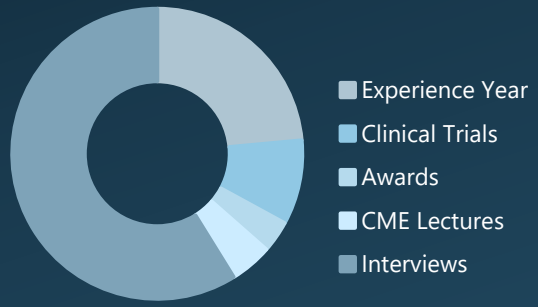
530 East 74th Street, New York, NY, 10021

Jae Park is a hematologist-oncologist specializing in leukemia, with a strong focus on cellular therapies like CAR-T cell therapy. He leads and co-leads multiple clinical trials, working on personalized and potentially curative treatments for leukemia patients.

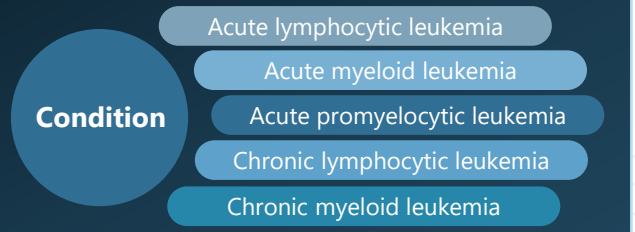
Memberships



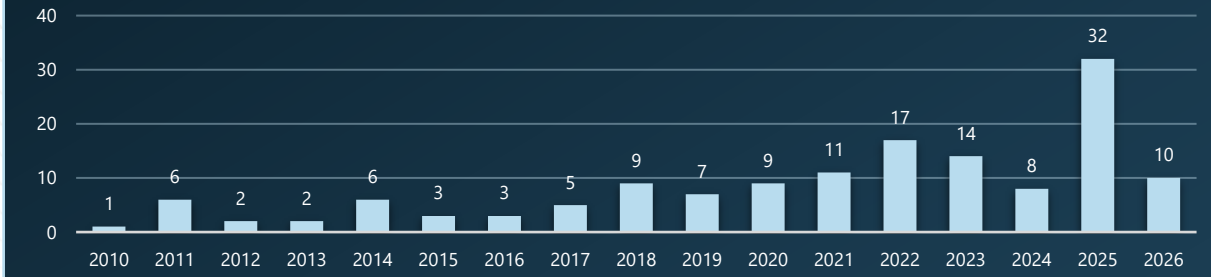
Records



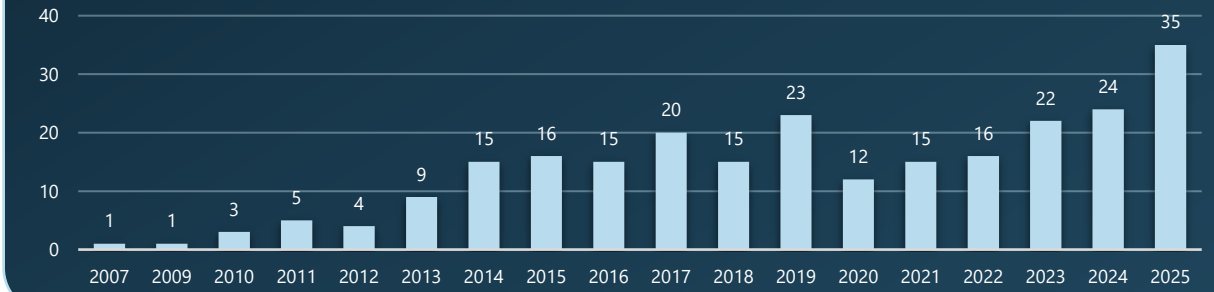
Condition He Treats



Publications



Meeting Abstracts



Research Activities



Industrial Relationships

Adaptive biotechnologies · Affyimmune Therapeutics · Allogene Therapeutics · Amgen · Ascentage Pharma · Autolus · artiva · CUROCELL · Synthekine · Bristol Myers Squibb · Iovance Biotherapeutics · Kite, A Gilead Company · Jazz Pharmaceuticals · Servier moved by you

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PAG DASHBOARD

About

MISSION
Drive clinical translation of cell & gene therapies worldwide.

VISION
Improve lives through safe and effective cell & gene therapies.

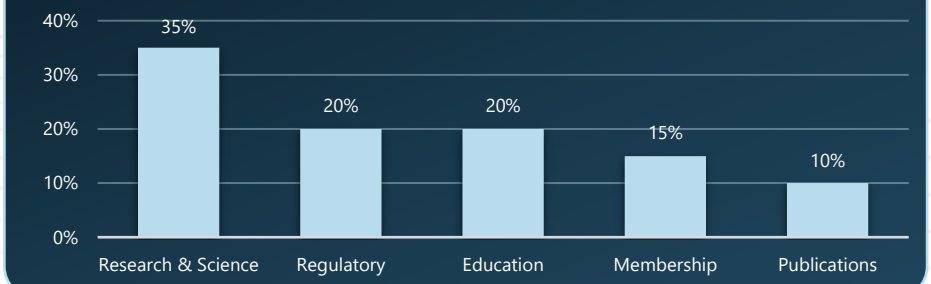
Value

1. **Innovation** — Promote translational research
2. **Excellence** — Strive for the highest standards
3. **Ethics** — Uphold the highest ethical standards
4. **Diversity** — Respect & support inclusivity
5. **Service** — Serve members & advocate for society

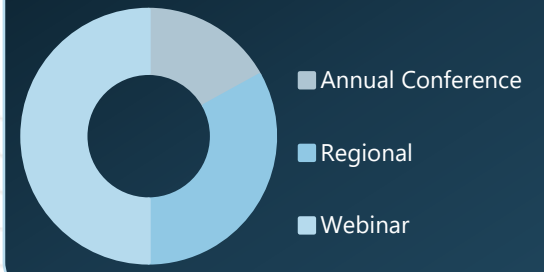
International Society ISCT Cell & Gene Therapy®

18K+	4.7K+	6	5
LinkedIn Followers	Twitter Followers	Events in 2026	Publications

ISCT Focus Area (Relative Engagement Weight)



Event Type Breakdown



Featured Event

ISCT 2026 — DUBLIN, IRELAND — MAY 6-9

650+	60+	250+	4-day
Abstracts	Countries Represented	Speakers	Curated Scientific Program

Annual Meetings - Global Locations (2027–2028)

ISCT 2027 North America

ISCT 2027
North America • Annual Meeting
May 12–15, 2027

ISCT 2028 Canada

ISCT 2028 CANADA
North America • Annual Meeting
May 31–June 3, 2028

Leadership



Queenie Jang
BSc, MBA
CEO
ISCT



Miguel Forte
MD, PhD
President
Kiji Therapeutics



Daniel J. Weiss
MD, PhD
President-Elect
University of Vermont



Jacques Galipeau
MD
Chair, SAC
University of Wisconsin in Madison

Partner Organizations

FACT — Foundation for the Accreditation of Cellular Therapy, at the University of Nebraska Medical Center
JACIE — Joint Accreditation Committee ISCT | EBMT
adbb — Association for the Advancement of Blood & Biotherapies
ABTCel-Gen
Junta de Andalucía
Alliance for Regenerative Medicine
ASTCT — American Society for Transplantation and Cellular Therapy
BAA — Biotherapeutics Association of Australasia
CARM
ICCBBA
CATAPULT — Cell and Gene Therapy
Cord Blood Association
CellCAN — Regenerative Medicine and Cell Therapy Network
eatris — European infrastructure for translational medicine
International Society for Extracellular Vesicles
Stem Cell Society Singapore
Standards Coordinating Body — Regenerative Medicine
The Korean Society of Blood and Marrow Transplantation

INSIGHTS CORNER

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Future Directions

01

CAR-T Therapy Expansion

CAR-T remains the key driver, with rapid shifts driven by allogeneic platforms and CRISPR innovations. Advancements include "armored" CAR-T cells that resist suppression, and inducible switches to control activation and minimize toxicity

02

Shift to Allogeneic "Off-the-Shelf" Therapies

Investors are fueling growth in allogeneic cell therapy platforms, enabling scalable, ready-to-use treatments that reduce patient wait times and expand access beyond autologous solutions

03

AI Integration

Generative AI and machine learning are reshaping drug discovery, trial design, and precision medicine accelerating breakthroughs and improving outcomes

04

Expanding Disease Indications and Combination Therapies

Researchers are working on solid tumor targeting to overcome immunosuppressive tumor microenvironments (TMEs) and improve T-cell infiltration
Integrating CAR-T with immune checkpoint inhibitors or monoclonal antibodies enhances therapeutic efficacy, broadening the treatment landscape in oncology

05

Sustainable and Cost-Efficient Manufacturing

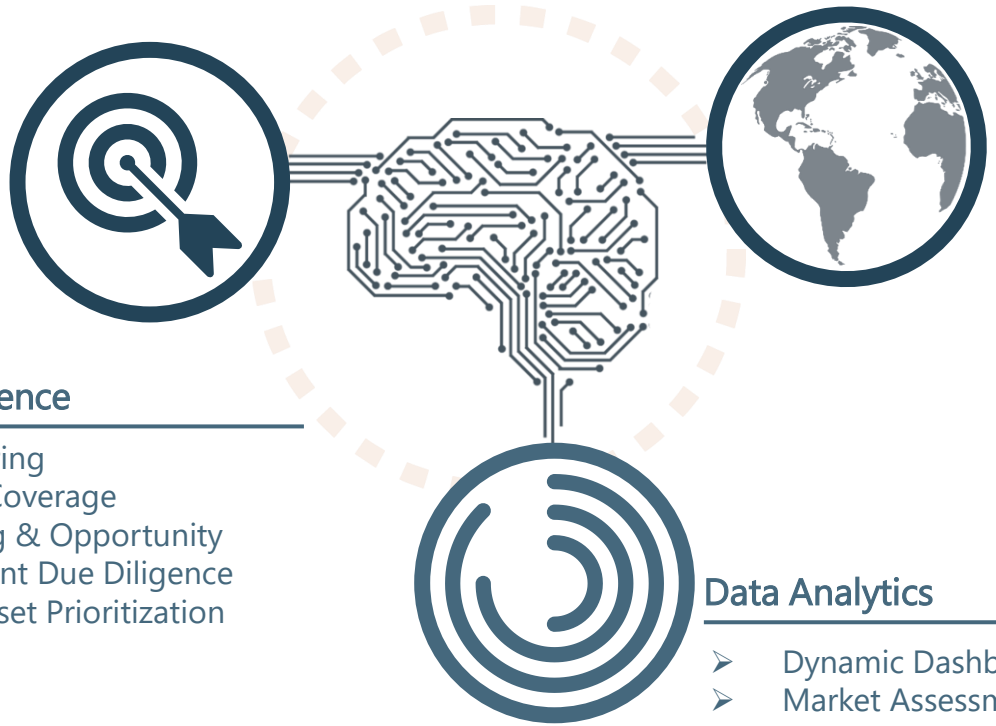
Adoption of automated bioreactors, closed-system production, and energy-efficient storage reducing treatment costs while boosting accessibility fueling global growth



This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

ABOUT COMPANY

Octavus delivers evidence-based consulting services to empower our clients in making well-informed decisions.



Market Research

- Quantitative & Qualitative Research
- Awareness, Trial & Usage (ATU) Studies
- KOL & PAG Identification & Mapping
- Concept/Product Testing
- AI & Human Confluence Transcription
- Content Analysis & Translation

Business Intelligence

- CI & Monitoring
- Conference Coverage
- Market Sizing & Opportunity
- Pre-investment Due Diligence
- Company/Asset Prioritization

Data Analytics

- Dynamic Dashboarding & Reporting
- Market Assessment & Epidemiology Analysis
- Sales Reporting & In-line Forecasting
- Social Media Analytics
- Clinical Analytics



OCTAVUS CONSULTING
Data.Insights.Decisions.

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